

```
In [547]: 1 import array
          2 import cmath
          3 import collections
          4 import copy
          5 import csv
          6 import datetime
          7 import decimal
          8 import functools
          9 import importlib
         10 import logging
         11 import math
         12 import pprint
         13 import random
         14 import re
         15 import statistics
         16 import string
         17 import sys
         18 import time
         19 import timeit
         20
         21 import pandas
```

```
In [2]: 1 num1 = 10_00_000
        2 print(num1)
```

```
1000000
```

```
In [3]: 1 print(format(0.1, ".25f"))
```

```
0.10000000000000000055511151
```

```
In [4]: 1 print(format(0.125, ".25f"))
```

```
0.12500000000000000000000000
```

```
In [5]: 1 print(0.1 + 0.1 + 0.1 == 0.3)
```

```
False
```

```
In [6]: 1 print(format(0.3, ".25f"))
```

```
0.2999999999999999888977698
```

```
In [7]: 1 print(format(0.2, ".25f"))
```

```
0.2000000000000000111022302
```

```
In [8]: 1 print(abs((0.1 + 0.1 + 0.1) - 0.3) < 0.001)
```

```
True
```

```
In [9]: 1 print((0.125).as_integer_ratio())
```

(1, 8)

```
In [10]: 1 print((10).      (25))
```

35

```
In [11]: 1 a = float(10)
        2 print(a)
```

10.0

```
In [12]: 1 c = (-4) ** 0.5
        2 print(c)
```

(1.2246467991473532e-16+2j)

```
In [13]: 1 print(type(c))
```

<class 'complex'>

```
In [14]: 1 print(c.real)
```

1.2246467991473532e-16

```
In [15]: 1 print(c.imag)
```

2.0

```
In [16]: 1 c = 10 + 2j
        2 print(type(c))
```

<class 'complex'>

```
In [17]: 1 print(c)
```

(10+2j)

```
In [18]: 1 try:
        2     print("2" < 2)
        3 except TypeError as ex:
        4     print(type(ex).      )
```

TypeError

```
In [19]: 1 print("2" == 2)
```

False

```
In [20]: 1 a = 0
        2 b = 1
        3 c = 500
        4 d = 0
        5 e = 1
        6 f = 500
        7 print(id(a), id(b), id(c), id(d), id(e), id(f))
```

11370064 11370096 128573173327504 11370064 11370096 128573173326608

```
In [21]: 1 print(0.1 * 3 - 0.3 < abs(1e-6))
```

True

```
In [22]: 1 a = 257
        2 print(a is 257)
```

False

<>:2: SyntaxWarning: "is" with 'int' literal. Did you mean "=="?

<>:2: SyntaxWarning: "is" with 'int' literal. Did you mean "=="?

/tmp/ipykernel_18021/1958631963.py:2: SyntaxWarning: "is" with 'int' literal. Did you mean "=="?
print(a is 257)

```
In [23]: 1 a = 10
        2 b = 0
        3 print(b != 0 and a / b)
```

False

```
In [24]: 1 a = 1e-6
        2 b = 2e-6
        3 tol = 1e-7
        4 print(abs(a - b) <= tol)
```

False

```
In [25]: 1 elapsed = 7835
        2 hours = elapsed // 3600
        3 minutes = (elapsed - (hours * 3600)) // 60
        4 seconds = elapsed - ((hours * 3600) + (minutes * 60))
        5 print(hours, minutes, seconds)
```

2 10 35

```
In [26]: 1 elapsed = 7835
        2 hours = elapsed // (60 * 60)
        3 minutes = (elapsed % (60 * 60)) // 60
        4 seconds = (elapsed % (60 * 60)) % 60
        5 print(hours, minutes, seconds)
```

2 10 35

```
In [27]: 1 a = None
        2 if a == None:
        3     a = "N/A"
        4 print(a)
```

N/A

```
In [28]: 1 a = None
        2 a = "N/A" if a is None else a
        3 print(a)
```

N/A

In [29]:

```
1 score = 700
2 rating = "Invalid"
3 if score < 0:
4     rating = "Negative"
5 elif score < 580:
6     rating = "Poor"
7 elif score < 670:
8     rating = "Fair"
9 elif score < 740:
10    rating = "Good"
11 elif score < 800:
12    rating = "Very Good"
13 elif score < 850:
14    rating = "Excellent"
15 print(rating)
```

Good

In [30]:

```
1 elapsed = 1000000
2 magnitude = "Invalid"
3 if elapsed < 60:
4     magnitude = "seconds"
5 elif elapsed < (60 * 60):
6     magnitude = "minutes"
7 elif elapsed < (60 * 60 * 24):
8     magnitude = "hours"
9 elif elapsed < (60 * 60 * 24 * 7):
10    magnitude = "days"
11 elif elapsed > (60 * 60 * 24 * 7):
12    magnitude = "weeks"
13 print(magnitude)
```

weeks

In [31]:

```
1 t = [1, 2], 3, 4
2 l = list(t)
3 print(l)
```

[[1, 2], 3, 4]

In [32]:

```
1 l[0][1] = 20
2 print(l)
```

[[1, 20], 3, 4]

In [33]:

```
1 print(t)
```

([1, 20], 3, 4)

In [34]:

```
1 for i in t:
2     print(id(i))
```

```
128573172499456
11370160
11370192
```

```
In [35]: 1 for i in l:
          2     print(id(i))
```

```
128573172499456
11370160
11370192
```

```
In [36]: 1 for i, j in zip(l, t):
          2     print(i is j)
```

```
True
True
True
```

```
In [37]: 1 s = "ABCDEFGHIIJ"
          2 print(s[3:6:-1])
```

```
In [38]: 1 print(s[3:5 + 1])
```

```
DEF
```

```
In [39]: 1 m1 = [[0, 0, 0], [0, 0, 0], [0, 0, 0]]
          2 m2 = m1[0:2]
          3 print(m1[0] is m2[1])
```

```
False
```

```
In [40]: 1 m1[0] = "Python"
          2 print(m1)
```

```
['Python', [0, 0, 0], [0, 0, 0]]
```

```
In [41]: 1 print(m2)
```

```
[[0, 0, 0], [0, 0, 0]]
```

```
In [42]: 1 m1[1][0] = 100
```

```
In [43]: 1 print(m1)
```

```
['Python', [100, 0, 0], [0, 0, 0]]
```

```
In [44]: 1 print(m2)
```

```
[[0, 0, 0], [100, 0, 0]]
```

```
In [45]: 1 print("abcdef"[-1:-4])
```

```
In [46]: 1 print("abcdef"[1:4:-1])
```

```
In [47]: 1 print("racecar" == "racecar"[::-1])
```

```
True
```

```
In [48]: 1 list1 = [0, 1, 2, 3, 4]
        2 list1[0:3] = "abc"
        3 print(list1)
```

```
['a', 'b', 'c', 3, 4]
```

```
In [49]: 1 list1[0:3] = ["a", "b", "c"]
        2 print(list1)
```

```
['a', 'b', 'c', 3, 4]
```

```
In [50]: 1 list1[0:3] = ["a", "b"]
        2 print(list1)
```

```
['a', 'b', 3, 4]
```

```
In [51]: 1 del list1[2]
        2 print(list1)
```

```
['a', 'b', 4]
```

```
In [52]: 1 del list1[0:2]
        2 print(list1)
```

```
[4]
```

```
In [53]: 1 list1.extend("ABC")
        2 print(list1)
```

```
[4, 'A', 'B', 'C']
```

```
In [54]: 1 list1.insert(0, "YO!")
        2 print(list1)
```

```
['YO!', 4, 'A', 'B', 'C']
```

```
In [55]: 1 list1[:-3:-1] = "cb"
        2 print(list1)
```

```
['YO!', 4, 'A', 'b', 'c']
```

```
In [56]: 1 l = []
        2 print(timeit.timeit("l.append(1)", globals = globals(), number = 100_000))
```

```
0.002112314999976661
```

```
In [57]: 1 print(len(l))
```

```
100000
```

```
In [58]: 1 list2 = []
        2 print(timeit.timeit("list2.insert(0, 1)", globals = globals(), number =
```

```
100_000))
```

```
1.5377864980000027
```

```
In [59]: 1 print(len(list2))
```

```
100000
```

```
In [60]: 1 print(1.249788 / 0.0018043)
```

```
692.6719503408524
```

```
In [61]: 1 list2 = []  
2 print(timeit.timeit("list2.insert(len(list2), 1)", globals = globals(), number  
= 100_000))
```

```
0.008461738999983481
```

```
In [62]: 1 print(0.0057634 / 0.0018043)
```

```
3.194258161059691
```

```
In [63]: 1 l1 = [[0, 1], 2, "3"]  
2 l2 = l1[:]  
3 for i in l1:  
4     print(id(i))
```

```
128573172569024  
11370128  
11434576
```

```
In [64]: 1 for i in l2:  
2     print(id(i))
```

```
128573172569024  
11370128  
11434576
```

```
In [65]: 1 l2[0][0] = "0"  
2 for i in l1:  
3     print(id(i))
```

```
128573172569024  
11370128  
11434576
```

```
In [66]: 1 for i in l2:  
2     print(id(i))
```

```
128573172569024  
11370128  
11434576
```

```
In [67]: 1 for i in l1:  
2     print(i)
```

```
['0', 1]  
2  
3
```

```
In [68]: 1 for i in l2:  
        2     print(i)
```

```
['0', 1]  
2  
3
```

```
In [69]: 1 l2[1] = "22"  
        2 for i in l1:  
        3     print(id(i))
```

```
128573172569024  
11370128  
11434576
```

```
In [70]: 1 for i in l2:  
        2     print(id(i))
```

```
128573172569024  
128574645937104  
11434576
```

```
In [71]: 1 for i in l1:  
        2     print(i)
```

```
['0', 1]  
2  
3
```

```
In [72]: 1 for i in l2:  
        2     print(i)
```

```
['0', 1]  
22  
3
```

```
In [73]: 1 l1 = [['0', 1], 2, "3"]  
        2 l2 = l1[:]  
        3 l2[0] = ["0", 1]  
        4 for i in l1:  
        5     print(id(i))
```

```
128573172599040  
11370128  
11434576
```

```
In [74]: 1 for i in l2:  
        2     print(id(i))
```

```
128573172596864  
11370128  
11434576
```

```
In [75]: 1 for i in l1:
         2     print(i)
```

```
[0, 1]
2
3
```

```
In [76]: 1 for i in l2:
         2     print(i)
```

```
['0', 1]
2
3
```

```
In [77]: 1 l1 = [[0, 1], 2, "3"]
         2 l2 = l1[:]
         3 print(id(l1), id(l2))
```

```
128573172599104 128573172598912
```

```
In [78]: 1 for i, j in zip(l1, l2):
         2     print(id(i), id(j))
```

```
128573172598848 128573172598848
11370128 11370128
11434576 11434576
```

```
In [79]: 1 for i, j in zip(l1[0], l2[0]):
         2     print(id(i), id(j))
```

```
11370064 11370064
11370096 11370096
```

```
In [80]: 1 l1 = [[0, 1], 2, "3"]
         2 l2 = l1.copy()
         3 print(id(l1), id(l2))
```

```
128573172597248 128573172597312
```

```
In [81]: 1 for i, j in zip(l1, l2):
         2     print(id(i), id(j))
```

```
128573172601472 128573172601472
11370128 11370128
11434576 11434576
```

```
In [82]: 1 for i, j in zip(l1[0], l2[0]):
         2     print(id(i), id(j))
```

```
11370064 11370064
11370096 11370096
```

```
In [83]: 1 l1 = [[0, 1], 2, "3"]
         2 l2 = copy.deepcopy(l1)
         3 print(id(l1), id(l2))
```

```
128574051537216 128573172602368
```

```
In [84]: 1 for i, j in zip(l1, l2):  
        2     print(id(i), id(j))
```

```
128573172600640 128573172437184  
11370128 11370128  
11434576 11434576
```

```
In [85]: 1 for i, j in zip(l1[0], l2[0]):  
        2     print(id(i), id(j))
```

```
11370064 11370064  
11370096 11370096
```

```
In [86]: 1 t1 = (0, "1", "23")  
        2 t2 = t1[:]  
        3 print(t1 is t2)
```

```
True
```

```
In [87]: 1 a, b, c, d = "abcd"  
        2 print(a, b, c, d)
```

```
a b c d
```

```
In [88]: 1 s = "FfEeDdCcBbAa"  
        2 lowerLetters = s[::-2]  
        3 upperLetters = s[-2::-2]  
        4 print(lowerLetters)
```

```
abcdef
```

```
In [89]: 1 print(upperLetters)
```

```
ABCDEF
```

```
In [90]: 1 t1 = 1, 2, 3, 4, 5, 6  
        2 t2 = 7, 8, 9, 10  
        3 t3 = 11, 12, 13, 14, 15, 16, 17  
        4 t = t1 + t2 + t3  
        5 l = list(t)  
        6 l[::2] = [0] * len(l[::2])  
        7 print(tuple(l))
```

```
(0, 2, 0, 4, 0, 6, 0, 8, 0, 10, 0, 12, 0, 14, 0, 16, 0)
```

```
In [91]: 1 m = [[0, 0, 0], [0, 0, 0], [0, 0, 0]]  
        2 m[0][0] = 1  
        3 m[1][1] = 1  
        4 m[2][2] = 1  
        5 print(m)
```

```
[[1, 0, 0], [0, 1, 0], [0, 0, 1]]
```

```
In [92]: 1 m = [[0, 0, 0], [0, 0, 0], [0, 0, 0]]
2 m1 = copy.deepcopy(m)
3 m1[0][0] = 1
4 m1[1][1] = 1
5 m1[2][2] = 1
6 print(m1)
```

```
[[1, 0, 0], [0, 1, 0], [0, 0, 1]]
```

```
In [93]: 1 print(m)
2 data = [(100, "USD", "EUR", 0.83), (100, "USD", "CAD", 1.27), (100, "CAD",
3 "EUR", 0.65)]
4 USDAmountInEUR = data[0][0] * data[0][3]
5 print(str(data[0][0]) + " " + data[0][1] + " = " + str(USDAmountInEUR) + " " +
str(data[0][2]))
```

```
[[0, 0, 0], [0, 0, 0], [0, 0, 0]]
100 USD = 83.0 EUR
```

```
In [94]: 1 USDAmountInCAD = data[1][0] * data[1][3]
2 print(str(data[1][0]) + " " + data[1][1] + " = " + str(USDAmountInCAD) + " " +
str(data[1][2]))
```

```
100 USD = 127.0 CAD
```

```
In [95]: 1 CADAmountInEUR = data[2][0] * data[2][3]
2 print(str(data[2][0]) + " " + data[2][1] + " = " + str(CADAmountInEUR) + " " +
str(data[2][2]))
```

```
100 CAD = 65.0 EUR
```

```
In [96]: 1 print(ord("A"))
```

```
65
```

```
In [97]: 1 print("\N{SNAKE}")
```

```
🐍
```

```
In [98]: 1 print(ord("\N{SNAKE}"))
```

```
128013
```

```
In [99]: 1 print(hex(ord("\N{SNAKE}"))) 
```

```
0x1f40d
```

```
In [100]: 1 print("\u03b1")
```

```
α
```

```
In [101]: 1 print("\U0001f40d")
```

```
🐍
```

```
In [102]: 1 print(int("1f40d", 16))
```

```
128013
```

```
In [103]: 1 print("\N{SNAKE} is a snake")
```

```
🐍 is a snake
```

```
In [104]: 1 print(int("7ff", 16))
```

```
2047
```

```
In [105]: 1 print(int("ffff", 16))
```

```
65535
```

```
In [106]: 1 print(int("10ffff", 16))
```

```
1114111
```

```
In [107]: 1 print(bin(2047))
```

```
0b1111111111
```

```
In [108]: 1 print(len(bin(2047)))
```

```
13
```

```
In [109]: 1 print(bin(65535))
```

```
0b1111111111111111
```

```
In [110]: 1 print(len(bin(65535)))
```

```
18
```

```
In [111]: 1 print(bin(1114111))
```

```
0b10000111111111111111
```

```
In [112]: 1 print(len(bin(1114111)))
```

```
23
```

```
In [113]: 1 print(bin(128013))
```

```
0b111111010000001101
```

```
In [114]: 1 print("\n{snake}")
```

```
{snake}
```

```
In [115]: 1 a = True  
2 b = True  
3 print(a is b)
```

```
True
```

```
In [116]: 1 a = False
          2 b = False
          3 print(a is b)
```

True

```
In [117]: 1 a = True
          2 print(id(a))
```

11120192

```
In [118]: 1 a = False
          2 print(id(a))
```

11045056

```
In [119]: 1 b = False
          2 print(id(b))
```

11045056

```
In [120]: 1 ?
```

Docstring:

Return the lowest index in S where substring sub is found, such that sub is contained within S[start:end].

Optional arguments start and end are interpreted as in slice notation.
Return -1 on failure.

Type: method_descriptor

```
In [121]: 1 print(f"1 + 1 = {1 + 1}")
```

1 + 1 = 2

```
In [122]: 1 bid = 1.5760
          2 ask = 1.5763
          3 print("bid: {}, ask: {}, spread: {}".format(bid, ask, ask - bid))
```

bid: 1.576, ask: 1.5763, spread: 0.000299999999999996696

```
In [123]: 1 print("bid: {:.4f}, ask: {:.4f}, spread: {:.4f}".format(bid, ask, ask - bid))
```

bid: 1.5760, ask: 1.5763, spread: 0.0003

```
In [124]: 1 print(f"bid: {bid:.4f}, ask: {ask:.4f}, spread: {ask - bid:.4f}")
```

bid: 1.5760, ask: 1.5763, spread: 0.0003

```
In [125]: 1 print(f"bid: {bid:.4f}, ask: {ask:.4f}, spread: {(ask - bid):.4f}")
```

bid: 1.5760, ask: 1.5763, spread: 0.0003

```
In [126]: 1 string1 = "β, β, #"
          2 string1 = "Π, ú, θ, ω, ν"
```

```
3 chars = string1.split(", ")
4 print(chars)
```

```
['Π', 'ó', 'θ', 'ω', 'ν']
```

```
In [127]: 1 list1 = []
          2 for c in chars:
          3     list1.append(hex(ord(c)))
          4 print(list1)
```

```
['0x3a0', '0x3cd', '0x3b8', '0x3c9', '0x3bd']
```

```
In [128]: 1 list1 = []
          2 for c in chars:
          3     list1.append(c.upper())
          4 print(list1)
```

```
['Π', 'Υ', 'Θ', 'Ω', 'Ν']
```

```
In [129]: 1 list1 = []
          2 for c in chars:
          3     list1.append(c.lower())
          4 print(list1)
```

```
['π', 'ó', 'θ', 'ω', 'ν']
```

```
In [130]: 1 a = 42
          2 print(f"The number {a} is not even") if a % 2 else print(f"The number {a} is even")
```

```
The number 42 is even
```

```
In [131]: 1 a = 41
          2 print(f"The number {a} is not even") if a % 2 else print(f"The number {a} is even")
```

```
The number 41 is not even
```

```
In [132]: 1 a = 42
          2 print("The number {} is not even".format(a)) if a % 2 else print("The number {} is even".format(a))
```

```
The number 42 is even
```

```
In [133]: 1 a = 41
          2 print("The number {} is not even".format(a)) if a % 2 else print("The number {} is even".format(a))
```

```
The number 41 is not even
```

```
In [134]: 1 a = 0.3
          2 b = 0.1
          3 print(f"{a} / {b} = {a / b:.4f}")
```

```
0.3 / 0.1 = 3.0000
```

```
In [135]: 1 a = 42
          2 qualifier = "is not" if a % 2 else "is"
          3 print(f"The number {a} {qualifier} even")
```

The number 42 is even

```
In [136]: 1 a = 421
          2 qualifier = "is not" if a % 2 else "is"
          3 print(f"The number {a} {qualifier} even")
```

The number 421 is not even

```
In [137]: 1 print(range(10))
```

range(0, 10)

```
In [138]: 1 print(type(range(10)))
```

<class 'range'>

```
In [139]: 1 for i in range(3):
          2     print(i)
          3     print(i)
```

0
1
2
2

```
In [140]: 1 for i in "abc":
          2     pass
          3     print(i)
```

c

```
In [141]: 1 raggedArray = [[1, 2], [3, 4, 5, 6], [7, 8, 9], [10]]
          2 for r in range(len(raggedArray)):
          3     for c in range(len(raggedArray[r])):
          4         print(f"[{r}, {c}] = {raggedArray[r][c]}")
          5     print("-" * 5)
```

[0, 0] = 1
[0, 1] = 2

[1, 0] = 3
[1, 1] = 4
[1, 2] = 5
[1, 3] = 6

[2, 0] = 7
[2, 1] = 8
[2, 2] = 9

[3, 0] = 10

```
In [142]: 1 identityMatrix = []
          2 size = 10
          3 for r in range(size):
          4     row = []
          5     for c in range(size):
          6         if r == c:
          7             row.append(1)
          8         else:
          9             row.append(0)
         10     identityMatrix.append(row)
         11 identityMatrix
```

```
Out[142]: [[1, 0, 0, 0, 0, 0, 0, 0, 0, 0],
           [0, 1, 0, 0, 0, 0, 0, 0, 0, 0],
           [0, 0, 1, 0, 0, 0, 0, 0, 0, 0],
           [0, 0, 0, 1, 0, 0, 0, 0, 0, 0],
           [0, 0, 0, 0, 1, 0, 0, 0, 0, 0],
           [0, 0, 0, 0, 0, 1, 0, 0, 0, 0],
           [0, 0, 0, 0, 0, 0, 1, 0, 0, 0],
           [0, 0, 0, 0, 0, 0, 0, 1, 0, 0],
           [0, 0, 0, 0, 0, 0, 0, 0, 1, 0],
           [0, 0, 0, 0, 0, 0, 0, 0, 0, 1]]
```

```
In [143]: 1 raggedArray = [[1, 2], [3, 4, 5, 6], [7, 8, 9], [10]]
          2 for i, r in enumerate(raggedArray):
          3     for j, e in enumerate(raggedArray[i]): # e for element, cuz it's not a
          4         column technically
          5             print(f"[{i}, {j}] = {raggedArray[i][j]}")
             print("-" * 8)
```

```
[0, 0] = 1
[0, 1] = 2
-----
[1, 0] = 3
[1, 1] = 4
[1, 2] = 5
[1, 3] = 6
-----
[2, 0] = 7
[2, 1] = 8
[2, 2] = 9
-----
[3, 0] = 10
-----
```

```
In [144]: 1 print(f"{0.1:.4f}")
```

```
0.1000
```

```
In [145]: 1 print("{:.4f}".format(0.1))
```

```
0.1000
```

```
In [146]: 1 total, count = (0,) * 2
          2 data = [10.5, 11.2, 9.8, None, 11.5, None]
          3 for e in data:
```

```
4         if e:
5             total += e
6             count += 1
7     average = total / count
8     print(average)
```

10.75

```
In [147]: 1 for i, e in enumerate(data):
2         if not e:
3             data[i] = average
4         print(data)
```

[10.5, 11.2, 9.8, 10.75, 11.5, 10.75]

```
In [148]: 1 data = [10.5, 11.2, 9.8, None, 11.5, None]
2     count = sum(1 for e in data if e)
3     total = sum(e for e in data if e)
4     average = total / count
5     print(sum, average)
```

<built-in function sum> 10.75

```
In [149]: 1 data = [10.5, 11.2, 9.8, None, 11.5, None]
2     average = statistics.fmean(e for e in data if e)
3     data = [average if e is None else e for e in data]
4     print(data)
```

[10.5, 11.2, 9.8, 10.75, 11.5, 10.75]

```
In [150]: 1 data = [10.5, 11.2, 9.8, None, 11.5, None]
2     df = pandas.DataFrame(data)
3     df.fillna(df.mean(), inplace = True)
4     print(df[0].tolist())
```

[10.5, 11.2, 9.8, 10.75, 11.5, 10.75]

```
In [151]: 1 print(len(str((1, 2))))
```

6

```
In [152]: 1 print(str((1, 2)))
```

(1, 2)

```
In [153]: 1 m, n = 4, 5
2     for i in range(1, m):
3         for j in range(1, n):
4             print(f"{i} x {j} = {i * j}")
5     print("-" * 8)
```

```
1 x 1 = 1
1 x 2 = 2
1 x 3 = 3
1 x 4 = 4
-----
2 x 1 = 2
2 x 2 = 4
2 x 3 = 6
2 x 4 = 8
-----
3 x 1 = 3
3 x 2 = 6
3 x 3 = 9
3 x 4 = 12
-----
```

In [154]:

```
1
2
3 data = ("2021-01-01", 10, 20], ["2021-01-02", 20, 18], ["2021-01-03", -10,
4 10], ["2021-01-04", 100, 102], ["2021-01-05", 20, 45])
5 largest = 0
6 for e in data:
7     distance = abs(e[2] - e[1])
8     e.append(distance)
9     if distance > largest:
10         dateOfLargest = e[0]
11         largest = distance
12 print(data)
13 print(dateOfLargest)
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```

```
In [156]: 1 tuple1 = ([0],)
          2 print(type(tuple1))
          3 print(tuple1)
          4 print(id(tuple1))
          5 print(id(tuple1[0]))
          6 tuple1[0][0] += 1
          7 print(tuple1)
          8 print(id(tuple1))
          9 print(id(tuple1[0]))
```

```
<class 'tuple'>
([0],)
128573173295168
128573172611328
([1],)
128573173295168
128573172611328
```

```
In [157]: 1 print(hash(True))
          2 a = True
          3 print(hash(a))
          4 b = True
          5 print(hash(b))
          6 a = b
          7 print(hash(a))
```

```
1
1
1
1
```

```
In [158]: 1 print(hash(257))
          2 a = 257
          3 print(hash(a))
          4 b = 257
          5 print(hash(b))
          6 a = b
          7 print(hash(a))
```

```
257
257
257
257
```

```
In [159]: 1 print(hash(2570))
          2 a = 2570
          3 print(hash(a))
          4 b = 2570
          5 print(hash(b))
          6 a = b
          7 print(hash(a))
```

```
2570
2570
2570
2570
```

```
In [160]: 1 print(hash(25))
          2 a = 25
          3 print(hash(a))
          4 b = 25
          5 print(hash(b))
          6 a = b
          7 print(hash(a))
```

```
25
25
25
25
```

```
In [161]: 1 print(hash(2.5))
          2 a = 2.5
          3 print(hash(a))
          4 b = 2.5
          5 print(hash(b))
          6 print(hex(hash(b)))
          7 a = b
          8 print(len(hex(hash(a))))
          9 print(hash(b))
```

```
1152921504606846978
1152921504606846978
1152921504606846978
0x10000000000000002
18
1152921504606846978
```

```
In [162]: 1 for i in range(5):
          2     for j in range(4):
          3         pass
          4 print(i, j)
```

```
4 3
```

```
In [163]: 1 s1 = """
          2 "'And' and 'or' are the basic operations of logic. Together with 'no' (the
          3 logical operation
          4 of negation) they are a complete set of basic logical operations – all other
          5 logical
          6 operations, no matter how complex, can be obtained by suitable combinations of
          7 these."
          8 — John von Neumann, The Computer and the Brain
          9 """
          10
          11 s2 = """
          12 Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor
          13 incididunt ut
```

```

7 labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud
exercitation ullamco laboris
8 nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in
reprehenderit in voluptate velit
9 esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat
1 non proident, sunt
0 in culpa qui officia deserunt mollit anim id est laborum.
1 """
1
1 charCounts = {}
2 for c in s1:
1     if (ord(c) >= ord("A") and ord(c) <= ord("Z")) or (ord(c) >= ord("a")
3 and ord(c) <= ord("z")):
1         charCounts[c] = charCounts.get(c, 0) + 1
4 print(charCounts)
1
5
1
6
1
7
1
8
1
9
2
0

```

```

{'A': 1, 'n': 20, 'd': 4, 'a': 23, 'o': 31, 'r': 12, 'e': 27, 't': 21, 'h': 11, 'b': 7, 's': 9, 'i':
17, 'c': 10, 'p': 7, 'f': 4, 'l': 12, 'g': 6, 'T': 2, 'w': 2, 'y': 2, 'm': 6, 'x': 1, 'u': 3, 'J':
1, 'v': 1, 'N': 1, 'C': 1, 'B': 1}

```

```

In [164]: 1 d1 = {"a": 10, "b": 20, "c": 30}
2 d2 = {"d": 100, "e": 200, "f": 300}
3 d3 = {"f": 30, "g": 40}
4 keysList, valuesList = [], []
5 for d in d1, d2, d3:
6     for k, v in d.items():
7         keysList.append(k)
8         valuesList.append(v)
9 print(keysList)
10 print(valuesList)

```

```

['a', 'b', 'c', 'd', 'e', 'f', 'f', 'g']
[10, 20, 30, 100, 200, 300, 30, 40]

```

```

In [165]: 1 keysList, valuesList = [], []
2 keysList = list(d1.keys())
3 keysList.extend(list(d2.keys()))
4 keysList.extend(list(d3.keys()))
5 print(keysList)

```

```

['a', 'b', 'c', 'd', 'e', 'f', 'f', 'g']

```

```
In [166]: 1 valuesList = list(d1.values())
          2 valuesList.extend(list(d2.values()))
          3 valuesList.extend(list(d3.values()))
          4 print(valuesList)
```

```
[10, 20, 30, 100, 200, 300, 30, 40]
```

```
In [167]: 1 print(list(d1.keys()) + list(d2.keys()) + list(d3.keys()))
          2 print(list(d1.values()) + list(d2.values()) + list(d3.values()))
```

```
['a', 'b', 'c', 'd', 'e', 'f', 'f', 'g']
[10, 20, 30, 100, 200, 300, 30, 40]
```

```
In [168]: 1 grades = {"John": [90, 95, 98], "Eric": [86, 84, 92], "Michael": [90, 89, 85]}
          2 exam = {"Eric": 99, "John": 100}
          3 for k, v in grades.items():
          4     if k in exam:
          5         v.insert(0, exam[k])
          6     else:
          7         v.insert(0, None)
          8 print(grades)
```

```
{'John': [100, 90, 95, 98], 'Eric': [99, 86, 84, 92], 'Michael': [None, 90, 89, 85]}
```

```
In [169]: 1 for k, v in grades.items():
          2     score = exam.get(k)
          3     grades[k] = [score] + grades[k]
          4 print(grades)
```

```
{'John': [100, 100, 90, 95, 98], 'Eric': [99, 99, 86, 84, 92], 'Michael': [None, None, 90, 89, 85]}
```

```
In [170]: 1 for k, v in grades.items():
          2     v.insert(0, exam.get(k))
          3 print(grades)
```

```
{'John': [100, 100, 100, 90, 95, 98], 'Eric': [99, 99, 99, 86, 84, 92], 'Michael': [None, None, None, 90, 89, 85]}
```

```
In [171]: 1 for k in grades:
          2     print(k)
```

```
John
Eric
Michael
```

```
In [172]: 1 l = ["AAPL", "AAPL", "Aapl", "aapl", "MSFT"]
          2 print(list(set(l)))
```

```
['AAPL', 'Aapl', 'MSFT', 'aapl']
```

```
In [173]: 1 l = [e.casefold() for e in l]
          2 print(list(set(l)))
```

```
['aapl', 'msft']
```

```
In [174]: 1 se = {e.casefold() for e in l}
          2 print(se)
```

```
{'aapl', 'msft'}
```

```
In [175]: 1 data = {"d1": {"a": 1, "b": 2, "c": 3}, "d2": {"b": 20, "c": 30, "d": 40},
          2 "d3": {"d": 100, "x": 200}}
          3 uniqueKeys = {k for v in (data["d1"], data["d2"], data["d3"]) for k in v}
          4 print(uniqueKeys)
```

```
{'d', 'x', 'c', 'b', 'a'}
```

```
In [176]: 1 try:
          2     s1 = {[0, 1]}
          3     print(s1)
          4 except TypeError as ex:
          5     print(type(ex)., ":", " ", *ex.args, ".", sep = " ")
```

```
TypeError: cannot use 'list' as a set element (unhashable type: 'list').
```

```
In [177]: 1 m = [0] * 3
          2 print(m)
          3 print(m[0] is m[1])
```

```
[0, 0, 0]
True
```

```
In [178]: 1 m = [[0] * 3 for _ in range(3)]
          2 print(m)
          3 print(m[0] is m[1])
          4 print(m[0][0] is m[0][1])
```

```
[[0, 0, 0], [0, 0, 0], [0, 0, 0]]
False
True
```

```
In [179]: 1 m = [[10000 for i in range(3)] for _ in range(4)]
          2 print(m)
```

```
[[10000, 10000, 10000], [10000, 10000, 10000], [10000, 10000, 10000], [10000, 10000, 10000]]
```

```
In [180]: 1 pprint.pprint(m, width = 1)
```

```
[[10000,
  10000,
  10000],
 [10000,
  10000,
  10000],
 [10000,
  10000,
  10000],
 [10000,
  10000,
  10000]]
```

```
In [181]: 1 print(m[0] is m[1])
          2 print(m[0][0] is m[0][1])
          3 print(hash(m[0][0]), hash(m[0][1]))
```

```
False
True
10000 10000
```

```
In [182]: 1 print(id(m[0][0]), id(m[0][1]))
```

```
128573172653872 128573172653872
```

```
In [183]: 1 m[0][0] = 1001
          2 pprint.pprint(m)
```

```
[[1001, 10000, 10000],
 [10000, 10000, 10000],
 [10000, 10000, 10000],
 [10000, 10000, 10000]]
```

```
In [184]: 1 print(id(m[0][0]), id(m[0][1]))
          2 print(m[0][0] is m[0][1])
```

```
128573172653712 128573172653872
False
```

```
In [185]: data = [{"open": 100, "high": 120, "low": 90, "close": 110}, {"open": 110,
1 "high": 130, "low": 80, "close": 120}, {"open": 120, "high": 140, "low": 70,
2 "close": 130}, {"open": 130, "high": 150, "low": 60, "close": 140},]
3 ranges = []
4 for d in data:
5     ranges.append(d["high"] - d["low"])
print(ranges)
```

```
[30, 50, 70, 90]
```

```
In [186]: data = [{"open": 100, "high": 120, "low": 90, "close": 110}, {"open": 110,
1 "high": 130, "low": 80, "close": 120}, {"open": 120, "high": 140, "low": 70,
2 "close": 130}, {"open": 130, "high": 150, "low": 60, "close": 140},]
3 ranges = [d["high"] - d["low"] for d in data]
print(ranges)
```

```
[30, 50, 70, 90]
```

```
In [187]: 1 bogusComposites = {i for i in range(1, 101) for j in range(2, 10) if i % j ==
          2 0}
          3 allNumbers = {i for i in range(11, 101)} | {1}
          print(allNumbers - bogusComposites)
```

```
{1, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89, 97}
```

```
In [188]: 1 allNumbers = set(range(1, 101))
          2 print(allNumbers - bogusComposites)
```

```
{1, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89, 97}
```

In [189]:

```
data = [{"symbol": "ABCD", "name": "ABCD Company", "ranking": 2, "risk": 0.2},
{"symbol": "BCDE", "name": "BCDE Company", "ranking": 5, "risk": 0.2},
{"symbol": "CDEF", "name": "CDEF Company", "ranking": 8, "risk": 0.5},
{"symbol": "DEFG", "name": "DEFG Company", "ranking": 7, "risk": 0.8},
{"symbol": "EFGH", "name": "EFGH Company", "ranking": 9, "risk": 0.6},
{"symbol": "FGHI", "name": "FGHI Company", "ranking": 10, "risk": 0.1},
{"symbol": "GHIJ", "name": "GHIJ Company", "ranking": 3, "risk": 0.6},
{"symbol": "HIJK", "name": "HIJK Company", "ranking": 5, "risk": 0.5},
{"symbol": "IJKL", "name": "IJKL Company", "ranking": 5, "risk": 0.7},
{"symbol": "JKLM", "name": "JKLM Company", "ranking": 6, "risk": 0.9},
{"symbol": "KLMN", "name": "KLMN Company", "ranking": 6, "risk": 0.4},
{"symbol": "LMNO", "name": "LMNO Company", "ranking": 8, "risk": 0.4},
{"symbol": "MNOP", "name": "MNOP Company", "ranking": 8, "risk": 0.2},
{"symbol": "NOPQ", "name": "NOPQ Company", "ranking": 1, "risk": 0.5},
{"symbol": "OPQR", "name": "OPQR Company", "ranking": 9, "risk": 0.2},
{"symbol": "PQRS", "name": "PQRS Company", "ranking": 10, "risk": 0.9},
{"symbol": "QRST", "name": "QRST Company", "ranking": 3, "risk": 0.4},
{"symbol": "RSTU", "name": "RSTU Company", "ranking": 7, "risk": 0.3},
{"symbol": "STUV", "name": "STUV Company", "ranking": 8, "risk": 0.1},
{"symbol": "TUVW", "name": "TUVW Company", "ranking": 7, "risk": 0.9}]
filtered = [d for d in data if d["ranking"] > 5 and d["risk"] < 0.6]
print(filtered)
```

```
[{'symbol': 'CDEF', 'name': 'CDEF Company', 'ranking': 8, 'risk': 0.5}, {'symbol': 'FGHI', 'name': 'FGHI Company', 'ranking': 10, 'risk': 0.1}, {'symbol': 'KLMN', 'name': 'KLMN Company', 'ranking': 6, 'risk': 0.4}, {'symbol': 'LMNO', 'name': 'LMNO Company', 'ranking': 8, 'risk': 0.4}, {'symbol': 'MNOP', 'name': 'MNOP Company', 'ranking': 8, 'risk': 0.2}, {'symbol': 'OPQR', 'name': 'OPQR Company', 'ranking': 9, 'risk': 0.2}, {'symbol': 'RSTU', 'name': 'RSTU Company', 'ranking': 7, 'risk': 0.3}, {'symbol': 'STUV', 'name': 'STUV Company', 'ranking': 8, 'risk': 0.1}]
```

In [190]:

```
1 for d in filtered:
2     d["weight"] = float(f"{d["ranking"] * d["risk"]:.2f}")
3 print(filtered)
```

```
[{'symbol': 'CDEF', 'name': 'CDEF Company', 'ranking': 8, 'risk': 0.5, 'weight': 4.0}, {'symbol': 'FGHI', 'name': 'FGHI Company', 'ranking': 10, 'risk': 0.1, 'weight': 1.0}, {'symbol': 'KLMN', 'name': 'KLMN Company', 'ranking': 6, 'risk': 0.4, 'weight': 2.4}, {'symbol': 'LMNO', 'name': 'LMNO Company', 'ranking': 8, 'risk': 0.4, 'weight': 3.2}, {'symbol': 'MNOP', 'name': 'MNOP Company', 'ranking': 8, 'risk': 0.2, 'weight': 1.6}, {'symbol': 'OPQR', 'name': 'OPQR Company', 'ranking': 9, 'risk': 0.2, 'weight': 1.8}, {'symbol': 'RSTU', 'name': 'RSTU Company', 'ranking': 7, 'risk': 0.3, 'weight': 2.1}, {'symbol': 'STUV', 'name': 'STUV Company', 'ranking': 8, 'risk': 0.1, 'weight': 0.8}]
```

In [191]:

```
1 weightedStocks = [dict(symbol = d["symbol"], weight = d["weight"]) for d in
2 filtered]
3 print(weightedStocks)
```

```
[{'symbol': 'CDEF', 'weight': 4.0}, {'symbol': 'FGHI', 'weight': 1.0}, {'symbol': 'KLMN', 'weight': 2.4}, {'symbol': 'LMNO', 'weight': 3.2}, {'symbol': 'MNOP', 'weight': 1.6}, {'symbol': 'OPQR', 'weight': 1.8}, {'symbol': 'RSTU', 'weight': 2.1}, {'symbol': 'STUV', 'weight': 0.8}]
```

In [192]:

```
1 weightedStocks = [dict(symbol = d["symbol"], weight = round(d["ranking"] *
2 d["risk"], 2)) for d in data if d["ranking"] >= 5 and d["risk"] < 0.6]
3 print(weightedStocks)
```

```
[{'symbol': 'BCDE', 'weight': 1.0}, {'symbol': 'CDEF', 'weight': 4.0}, {'symbol': 'FGHI', 'weight': 1.0}, {'symbol': 'HIJK', 'weight': 2.5}, {'symbol': 'KLMN', 'weight': 2.4}, {'symbol': 'LMNO', 'weight': 3.2}, {'symbol': 'MNOP', 'weight': 1.6}, {'symbol': 'OPQR', 'weight': 1.8}, {'symbol': 'RSTU', 'weight': 2.1}, {'symbol': 'STUV', 'weight': 0.8}]
```

```
In [193]: 1 weightedStocks = [{"symbol": d["symbol"], "weight": round(d["ranking"] *  
2 d["risk"], 2)} for d in data if d["ranking"] >= 5 and d["risk"] < 0.6]  
print(weightedStocks)
```

```
[{'symbol': 'BCDE', 'weight': 1.0}, {'symbol': 'CDEF', 'weight': 4.0}, {'symbol': 'FGHI', 'weight': 1.0}, {'symbol': 'HIJK', 'weight': 2.5}, {'symbol': 'KLMN', 'weight': 2.4}, {'symbol': 'LMNO', 'weight': 3.2}, {'symbol': 'MNOP', 'weight': 1.6}, {'symbol': 'OPQR', 'weight': 1.8}, {'symbol': 'RSTU', 'weight': 2.1}, {'symbol': 'STUV', 'weight': 0.8}]
```

```
In [194]: 1 m = [[0] * 2] * 3  
2 print(m)  
3 for r in range(len(m)):  
4     for c in range(len(m[r])):  
5         print(id(m[r][c]))
```

```
[[0, 0], [0, 0], [0, 0]]  
11370064  
11370064  
11370064  
11370064  
11370064  
11370064
```

```
In [195]: 1 m = [[{0}] * 2] * 3  
2 print(m)  
3 for r in range(len(m)):  
4     for c in range(len(m[r])):  
5         print(id(m[r][c]))
```

```
[[{0}, {0}], [{0}, {0}], [{0}, {0}]]  
128573172485248  
128573172485248  
128573172485248  
128573172485248  
128573172485248  
128573172485248
```

```
In [196]: 1 m[0][0] = 1  
2 for r in range(len(m)):  
3     for c in range(len(m[r])):  
4         print(id(m[r][c]))  
5 print(m)
```

```
11370096  
128573172485248  
11370096  
128573172485248  
11370096  
128573172485248  
[[1, {0}], [1, {0}], [1, {0}]]
```

```
In [197]: 1 m = [[{0} for _ in range(2)] for _ in range(3)]
2 print(m)
3 for r in range(len(m)):
4     for c in range(len(m[r])):
5         print(id(m[r][c]))
6 m[0][0] = 1
7 print(m)
```

```
[[{0}, {0}], [{0}, {0}], [{0}, {0}]]
128573172485696
128573172482336
128573172482112
128573172486144
128573172487040
128573172485024
[[1, {0}], [{0}, {0}], [{0}, {0}]]
```

```
In [198]: 1 m = [[{0} for _ in range(2)] for _ in range(3)]
2 print(m)
3 for r in range(len(m)):
4     for c in range(len(m[r])):
5         print(id(m[r][c]))
6 m[0][0] = {1}
7 print(m)
```

```
[[{0}, {0}], [{0}, {0}], [{0}, {0}]]
128573172485696
128573172483680
128573172487488
128573172486816
128573172486592
128573172488160
[[{1}, {0}], [{0}, {0}], [{0}, {0}]]
```

```
In [199]: 1 m = [[0 for _ in range(2)] for _ in range(3)]
2 for r in range(len(m)):
3     for c in range(len(m[r])):
4         print(id(m[r][c]))
```

```
11370064
11370064
11370064
11370064
11370064
11370064
```

```
In [200]: 1 m = [[10000 for _ in range(2)] for _ in range(3)]
2 for r in range(len(m)):
3     for c in range(len(m[r])):
4         print(id(m[r][c]))
5 m[0][0] = 1
6 print(m)
```

```
128573172653264
128573172653264
128573172653264
128573172653264
128573172653264
128573172653264
[[1, 10000], [10000, 10000], [10000, 10000]]
```

```
In [201]: 1 m = [{0} for _ in range(2)] for _ in range(3)
          2 for r in range(len(m)):
          3     for c in range(len(m[r])):
          4         print(id(m[r][c]))
```

```
128573172483680
128573172487488
128573172485248
128573172486144
128573172488160
128573172486368
```

```
In [202]: 1 # print(input("Text box displayed ?"))
```

```
In [203]: 1 a = 0
          2 b = complex(real = 3.0, imag = 4)
          3 print(repr(a))
          4 print(repr(b))
          5 print(b)
```

```
0
(3+4j)
(3+4j)
```

```
In [204]: 1 values = [4, 3, 1, 2]
          2 try:
          3     minimum = abs(values[0])
          4     for value in values[1:]:
          5         if abs(value) < minimum:
          6             minimum = abs(value)
          7 except IndexError:
          8     minimum = 0
          9     pass
         10 finally:
         11     print(f"Minimum is: {minimum}")
```

```
Minimum is: 1
```

```
In [205]: 1 values = []
          2
          3 try:
          4     minimum = abs(values[0])
          5     for value in values[1:]:
          6         if abs(value) < minimum:
          7             minimum = abs(value)
          8 except IndexError:
          9     minimum = 0
```

```
10     pass
11 finally:
12     print(f"Minimum is: {minimum}")
```

Minimum is: 0

```
In [206]: 1 try:
          2     raise ValueError("Custom message")
          3 except ValueError as ex:
          4     print(type(ex).      + ": " + ex.args[0] + ".")
```

ValueError: Custom message.

```
In [ ]: 1 l = [4, 3, 2, 1, 0]
        2 itr = iter(l)
        3 try:
        4     while True:
        5         print(next(itr))
        6 except StopIteration as ex:
        7     print(type(ex).      , ": ", *ex.args, ".", sep = "")
```

```
4
3
2
1
0
StopIteration: .
```

```
In [208]: 1 try:
          2     raise IndexError
          3 except IndexError as ex:
          4     print(type(ex).      , ": ", *ex.args, ".", sep = "")
```

IndexError: .

```
In [209]: 1 rangeObject = range(10)
          2 rangeIterator = iter(rangeObject)
          3 print(next(rangeIterator))
          4 print(next(rangeIterator))
          5 print(next(rangeIterator))
          6 print(list(rangeIterator))
```

```
0
1
2
[3, 4, 5, 6, 7, 8, 9]
```

```
In [210]: 1 rangeObj = range(5)
          2 print(list(rangeObj))
          3 print(list(rangeObj))
```

```
[0, 1, 2, 3, 4]
[0, 1, 2, 3, 4]
```

```
In [211]: 1 enumIterator = enumerate(string.ascii_lowercase)
          2 print(enumIterator)
```

```

3 l = list(enumIterator)
4 print(l)
5 l = list(enumIterator)
6 print(dir(type(iter([0]))))
7 print(l)
8 try:
9     print(next(enumIterator))
10 except StopIteration as ex:
11     print(type(ex).      , ": ", *ex.args, ".", sep = "")

```

```

<enumerate object at 0x74efc6458680>
[(0, 'a'), (1, 'b'), (2, 'c'), (3, 'd'), (4, 'e'), (5, 'f'), (6, 'g'), (7, 'h'), (8, 'i'), (9, 'j'),
(10, 'k'), (11, 'l'), (12, 'm'), (13, 'n'), (14, 'o'), (15, 'p'), (16, 'q'), (17, 'r'), (18, 's'),
(19, 't'), (20, 'u'), (21, 'v'), (22, 'w'), (23, 'x'), (24, 'y'), (25, 'z')]
['_class_', '__delattr__', '__dir__', '__doc__', '__eq__', '__format__', '__ge__',
'__getattr__', '__getstate__', '__gt__', '__hash__', '__init__', '__init_subclass__',
'__iter__', '__le__', '__length_hint__', '__lt__', '__ne__', '__new__', '__next__', '__reduce__',
'__reduce_ex__', '__repr__', '__setattr__', '__setstate__', '__sizeof__', '__str__',
'__subclasshook__']
[]
StopIteration: .

```

```

In [212]: 1 print(type(enumIterator))
          2 print(type(iter(enumIterator)))
          3 print(issubclass(type(iter([0])), type(enumIterator)))

```

```

<class 'enumerate'>
<class 'enumerate'>
False

```

```

In [213]: 1 print(iter(enumIterator) is enumIterator)

True

```

```

In [214]: 1 print(type(iter({0, 1, 2})))

<class 'set_iterator'>

```

```

In [215]: 1 print(type(iter({0: "0", 1: "1", "2": 2})))

<class 'dict_keyiterator'>

```

```

In [ ]: 1 l = [4, 3, 2, 1, 0]
        2 itr = iter(l)
        3 for i in itr:
        4     print(i)

```

```

4
3
2
1
0

```

```

In [217]: 1 try:
          2     print(next(l))
          3 except TypeError as ex:
          4     print(type(ex).      , ": ", *ex.args, ".", sep = "")

```

TypeError: 'list' object is not an iterator.

```
In [218]: 1 start = time.perf_counter()
2 squaresGenerator = (i ** 10 for i in range(1_000_000))
3 end = time.perf_counter()
4 print(f"Elapsed: {end - start}")
```

Elapsed: 0.00016390199999705146

```
In [219]: 1 start = time.perf_counter()
2 squaresListComp = [i ** 10 for i in range(1_000_000)]
3 end = time.perf_counter()
4 print(f"Elapsed: {end - start}")
```

Elapsed: 0.23438802800001213

```
In [220]: 1 print((7.16e-5 / 1.824e-1) ** -1)
2 print(105 / 9)
3 print(10.5e1 / 9e0)
4 print(99 / 1)
5 print(9.9e1 / 1e0) # 1 order of magnitude difference means a multiple of
between 10 and 100
```

2547.486033519553
11.666666666666666
11.666666666666666
99.0
99.0

```
In [221]: 1 print(999_999 ** 10)
```

999990000004499988000020999974800020999988000000449999900000001

```
In [222]: 1 print(squaresGenerator is iter(squaresGenerator))
```

True

```
In [223]: 1 l = [10, "abc", 3.14, True]
2 for i, e in enumerate(l):
3     print(f"{i}: {e}")
```

0: 10
1: abc
2: 3.14
3: True

```
In [224]: 1 def f(a, *b, x, **y):
2         print("Positional/Named:", a)
3         print("Variable positional:", b)
4         print("Named only:", x)
5         print("Variable named:", y)
6
7
8 f(0, 1, 2, x = 3, y = 4, z = 5)
```

```
Positional/Named: 0
Variable positional: (1, 2)
Named only: 3
Variable named: {'y': 4, 'z': 5}
```

```
In [225]: 1 # f(10, 2, x = 10, 20)
```

```
In [226]: 1 f(10, 2, x = 10, y = 20)
```

```
Positional/Named: 10
Variable positional: (2,)
Named only: 10
Variable named: {'y': 20}
```

```
In [227]: 1 f(10, z = 2, x = 10, y = 20)
```

```
Positional/Named: 10
Variable positional: ()
Named only: 10
Variable named: {'z': 2, 'y': 20}
```

```
In [228]: 1 f(10, x = 2, y = 10, z = 20)
```

```
Positional/Named: 10
Variable positional: ()
Named only: 2
Variable named: {'y': 10, 'z': 20}
```

```
In [229]: 1 # f(a = 10, 2, x = 10, y = 20)
```

```
In [230]: 1 f(x = 10, y = 20, z = 30, a = 10)
```

```
Positional/Named: 10
Variable positional: ()
Named only: 10
Variable named: {'y': 20, 'z': 30}
```

```
In [231]: 1 try:
2         f(10, a = 2, x = 10, y = 20)
3     except TypeError as ex:
4         print(type(ex).      )
```

```
TypeError
```

```
In [232]: 1 def f1(a, *, x, **y):
2         print(f"a: {a}, x: {x}, y: {y}")
```

```
In [233]: 1 try:
2         f1(10, a = 20, x = 2, y = 10)
3     except TypeError as ex:
4         print(type(ex).      )
```

```
TypeError
```

```
In [234]: 1 f1(a = 1, x = 2)
```

```
a: 1, x: 2, y: {}
```

```
In [235]: 1 print(max("cba"))
```

```
c
```

```
In [236]: 1 def f2(a, b):  
2         print(f"a: {a}; b: {b}")  
3  
4  
5 f2(2, 3)
```

```
a: 2; b: 3
```

```
In [237]: 1 f2(2, b = 3)
```

```
a: 2; b: 3
```

```
In [238]: 1 # f2(a = 2, 3)
```

```
In [239]: 1 f2(b = 1, a = 0)
```

```
a: 0; b: 1
```

```
In [240]: 1 # def f3(a = 0, b):  
2         # print(f"a: {a}; b: {b}")
```

```
In [241]: 1 def f3(a, b = 0):  
2         print(f"a: {a}; b: {b}")  
3  
4  
5 f3(-1, 1)
```

```
a: -1; b: 1
```

```
In [242]: 1 # f3(a = -1, 1)
```

```
In [243]: 1 f3(a = -1, b = 1)
```

```
a: -1; b: 1
```

```
In [244]: 1 def f4(a, b = 1, *, x):  
2         print(a, b, x)  
3  
4  
5 f4(0, 2, x = 3)
```

```
0 2 3
```

```
In [245]: 1 def f5(a, b = 1, *, x = 2, y):  
2         print(a, b, x, y)  
3  
4  
5 f5(0, 2, x = 3, y = 4)
```

0 2 3 4

```
In [246]: 1 def f6(**x):
          2     print(x)
          3
          4
          5 f6(a = 1, b = 2, c = 3)
```

{'a': 1, 'b': 2, 'c': 3}

```
In [247]: 1 # def f7(*, **x):
          2 #     print(x)
          3 #
          4 #
          5 # f7(a = 1, b = 2)
```

```
In [248]: 1 def f8(a, *, b):
          2     print(f"a: {a}; b: {b}")
          3
          4
          5 f8(b = 1, a = 0)
```

a: 0; b: 1

```
In [249]: 1 def toJSON(a, *, b, **c):
          2     print(a, b, c)
          3
          4
          5 toJSON(0, b = 3, c1 = 0, c2 = 1)
```

0 3 {'c1': 0, 'c2': 1}

```
In [250]: 1 def avg(a, *b):
          2     return (sum(b) + a) / (len(b) + 1)
          3
          4
          5 print(avg(0, 1, 2, 3, 4))
          6 print(avg(0))
          7 print(avg(0, 1))
```

2.0
0.0
0.5

```
In [251]: 1 def separator(b = 10, *, a = "-"):
          2     return a * b
          3
          4
          5 print(separator())
          6 print(separator(a = "A"))
          7 print(separator(5))
          8 print(separator(a = "B", b = 5))
          9 print(separator(5, a = "B"))
```

```
-----  
AAAAAAAAA  
-----  
BBBBBB  
BBBBBB
```

```
In [252]: 1 print(sep = "|", *("AB", "CD"))
```

```
AB|CD
```

```
In [253]: 1 # def f(a, *b, c):  
2         #     print(a, b, c)  
3         #  
4         #  
5         # f(c = 2, 0, 1, 2)
```

```
In [254]: 1 # def f(*a, b = 0):  
2         #     print(a, b)  
3         #  
4         #  
5         # f(b = 1, 0, 2)
```

```
In [255]: 1 try:  
2         f(b = 1, *(0, 2))  
3 except TypeError as ex:  
4         print(type(ex).      )
```

```
TypeError
```

```
In [256]: 1 # def f(a, *, b):  
2         #     print(a, b)  
3         #  
4         #  
5         # f(b = 1, 0)
```

```
In [257]: 1 # def f(a, b):  
2         #     print(a, b)  
3         #  
4         #  
5         # f(b = 1, 0)
```

```
In [258]: 1 # def f(a, b = 2):  
2         #     print(a, b)  
3         #  
4         #  
5         # f(b = 1, 0)
```

```
In [259]: 1 def f(a, b = 2):  
2         print(a, b)  
3
```

```
4
5 f(b = 1, *(0,))
```

```
0 1
```

```
In [260]: 1 anonymous = lambda iterable: len(set(iterable))
2          print(anonymous("abcd"))
3          print(anonymous("abab"))
4          try:
5              print(anonymous())
6          except TypeError as ex:
7              print(type(ex).      )
8          print(anonymous(""))
9          print(anonymous([0, 1, 2]))
10         print(anonymous([0]))
11         print(anonymous([]))
12         print(anonymous([1, 1, 2, 2, 3, 4]))
```

```
4
2
TypeError
0
3
1
0
4
```

```
In [261]: 1 s = "This is the first sentence. This is the scecond sentence. This is not the
2          fourth sentence, it is the third sentence."
3
4          def f(s = ""):
5              return dict(collections.Counter(s))
6
7
8          print(f(s))
9          print(f(""))
```

```
{'T': 3, 'h': 9, 'i': 10, 's': 13, ' ': 20, 't': 13, 'e': 17, 'f': 2, 'r': 3, 'n': 10, 'c': 6, 'l':
3, 'o': 3, 'd': 2, 'u': 1, ',': 1}
{}
```

```
In [262]: 1 s = "This is the first sentence. This is the scecond sentence. This is not the
2          fourth sentence, it is the third sentence."
3
4          def f(s = ""):
5              words = re.split("\\.| |", s)
6              for word in words:
7                  if word == "":
8                      words.remove(word)
9              frequencies = dict.fromkeys(set(words), 0)
10             for word in words:
11                 frequencies[word] += 1
12             print(frequencies)
```

```
7
8 f(s)
9
10
11
12
13
14
15
```

```
{'is': 4, 'the': 4, 'fourth': 1, 'third': 1, 'it': 1, 'not': 1, 'first': 1, 'second': 1, 'sentence': 4, 'This': 3}
```

```
In [263]: 1 def f(s = ""):
2         s = s.replace(".", "")
3         s = s.replace(",", "")
4         words = s.split()
5         return dict(collections.Counter(words))
6
7
8 print(f(s))
```

```
{'This': 3, 'is': 4, 'the': 4, 'first': 1, 'sentence': 4, 'second': 1, 'not': 1, 'fourth': 1, 'it': 1, 'third': 1}
```

```
In [264]: 1 print(round(1.812))
```

```
2
```

```
In [265]: 1 n, d = (0.3).as_integer_ratio()
2         print(n, d)
3         print(1 / 3)
```

```
5404319552844595 18014398509481984
0.3333333333333333
```

```
In [266]: 1 print(round(0.325, 2))
```

```
0.33
```

```
In [267]: 1 print(f"{0.325:.40f}")
```

```
0.3250000000000000111022302462515654042363
```

```
In [268]: 1 def toJSON(a, *, b, **c):
2         print(a, b, c)
3
```

```

4
5 try:
6     toJSONN(0, 1, 2, b = 3, c1 = 4, c2 = 5)
7 except TypeError as ex:
8     print(type(ex).

```

TypeError

```

In [269]: 1 def toJSONN(a, *, b, **c):
2         print(a, b, c)
3
4
5 toJSONN(0, b = 3, c1 = 4, c2 = 5)

```

0 3 {'c1': 4, 'c2': 5}

```

In [270]: 1 # def toJSONN(a, *):
2         #     print(a)
3         #
4         #
5         # toJSONN(0, 1, 2)

```

```

In [271]: 1 def toJSONN(a, *args):
2         print(a, args)
3
4
5 toJSONN(0, 1, 2)

```

0 (1, 2)

```

In [272]: 1 l = ["AB CD", "EF CC", "GH BC", "IJ AC", "KL CA", "MN JQ", "OP BA"]
2         print(sorted(l, key = lambda e: (e[3], e[4])))

```

['IJ AC', 'OP BA', 'GH BC', 'KL CA', 'EF CC', 'AB CD', 'MN JQ']

```

In [273]: 1 # print(sorted(l, key = lambda e: e[len(e) - i] for i in range(len(e))))
2         print(sorted(l, key = lambda e: e[len(e) - 1]))

```

['KL CA', 'OP BA', 'EF CC', 'GH BC', 'IJ AC', 'AB CD', 'MN JQ']

```

In [274]: 1 a = (1 for _ in range(5))
2         print(a)
3         print(lambda _: "ABCD")
4         print(sorted)
5         print(range)

```

<generator object <genexpr> at 0x74efc64a24d0>
 <function <lambda> at 0x74efc644b740>
 <built-in function sorted>
 <class 'range'>

```

In [275]: 1 def f(*a, b):
2         print(a, b)
3

```

```
4
5 f(b = 0, *(1, 2))
```

```
(1, 2) 0
```

```
In [276]: 1 def f(a, *b, c):
          2     print(a, b, c)
          3
          4
          5 f(0, c = 1, *(2, 3))
```

```
0 (2, 3) 1
```

```
In [277]: 1 print(statistics.fmean)
```

```
<function fmean at 0x74effaaa6a30>
```

```
In [278]: 1 print(dict([("a", 97), ("b", 98), ("c", 99)]))
```

```
{'a': 97, 'b': 98, 'c': 99}
```

```
In [279]: 1 widgets = [f"w{i}" for i in range(1, 21)]
          2 skus = [f"sku{i}" for i in range(1, len(widgets) + 1)]
          3 print(widgets)
          4 print(skus)
```

```
['w1', 'w2', 'w3', 'w4', 'w5', 'w6', 'w7', 'w8', 'w9', 'w10', 'w11', 'w12', 'w13', 'w14', 'w15',
'w16', 'w17', 'w18', 'w19', 'w20']
['sku1', 'sku2', 'sku3', 'sku4', 'sku5', 'sku6', 'sku7', 'sku8', 'sku9', 'sku10', 'sku11', 'sku12',
'sku13', 'sku14', 'sku15', 'sku16', 'sku17', 'sku18', 'sku19', 'sku20']
```

```
In [280]: 1 print({w: s for w, s in zip(widgets, skus)})
```

```
{'w1': 'sku1', 'w2': 'sku2', 'w3': 'sku3', 'w4': 'sku4', 'w5': 'sku5', 'w6': 'sku6', 'w7': 'sku7',
'w8': 'sku8', 'w9': 'sku9', 'w10': 'sku10', 'w11': 'sku11', 'w12': 'sku12', 'w13': 'sku13', 'w14':
'sku14', 'w15': 'sku15', 'w16': 'sku16', 'w17': 'sku17', 'w18': 'sku18', 'w19': 'sku19', 'w20':
'sku20'}
```

```
In [281]: 1 print(dict(zip(widgets, skus)))
```

```
{'w1': 'sku1', 'w2': 'sku2', 'w3': 'sku3', 'w4': 'sku4', 'w5': 'sku5', 'w6': 'sku6', 'w7': 'sku7',
'w8': 'sku8', 'w9': 'sku9', 'w10': 'sku10', 'w11': 'sku11', 'w12': 'sku12', 'w13': 'sku13', 'w14':
'sku14', 'w15': 'sku15', 'w16': 'sku16', 'w17': 'sku17', 'w18': 'sku18', 'w19': 'sku19', 'w20':
'sku20'}
```

```
In [282]: 1 suits = "shdc" # Spades, Hearts, Diamonds, Clubs
          2 ranks = list("23456789") + ["10", "J", "Q", "K", "A"]
          3 print(suits)
          4 print(ranks)
```

```
shdc
['2', '3', '4', '5', '6', '7', '8', '9', '10', 'J', 'Q', 'K', 'A']
```

```
In [283]: 1 print([(rank, suit) for rank in ranks] for suit in suits))
```

```
[('2', 's'), ('3', 's'), ('4', 's'), ('5', 's'), ('6', 's'), ('7', 's'), ('8', 's'), ('9', 's'), ('10', 's'), ('J', 's'), ('Q', 's'), ('K', 's'), ('A', 's')], [('2', 'h'), ('3', 'h'), ('4', 'h'), ('5', 'h'), ('6', 'h'), ('7', 'h'), ('8', 'h'), ('9', 'h'), ('10', 'h'), ('J', 'h'), ('Q', 'h'), ('K', 'h'), ('A', 'h')], [('2', 'd'), ('3', 'd'), ('4', 'd'), ('5', 'd'), ('6', 'd'), ('7', 'd'), ('8', 'd'), ('9', 'd'), ('10', 'd'), ('J', 'd'), ('Q', 'd'), ('K', 'd'), ('A', 'd')], [('2', 'c'), ('3', 'c'), ('4', 'c'), ('5', 'c'), ('6', 'c'), ('7', 'c'), ('8', 'c'), ('9', 'c'), ('10', 'c'), ('J', 'c'), ('Q', 'c'), ('K', 'c'), ('A', 'c')]]
```

In [284]:

1

```
pprint.pprint([(rank, suit) for rank in ranks] for suit in suits))
```

```

[('2', 's'),
 ('3', 's'),
 ('4', 's'),
 ('5', 's'),
 ('6', 's'),
 ('7', 's'),
 ('8', 's'),
 ('9', 's'),
 ('10', 's'),
 ('J', 's'),
 ('Q', 's'),
 ('K', 's'),
 ('A', 's')],
[('2', 'h'),
 ('3', 'h'),
 ('4', 'h'),
 ('5', 'h'),
 ('6', 'h'),
 ('7', 'h'),
 ('8', 'h'),
 ('9', 'h'),
 ('10', 'h'),
 ('J', 'h'),
 ('Q', 'h'),
 ('K', 'h'),
 ('A', 'h')],
[('2', 'd'),
 ('3', 'd'),
 ('4', 'd'),
 ('5', 'd'),
 ('6', 'd'),
 ('7', 'd'),
 ('8', 'd'),
 ('9', 'd'),
 ('10', 'd'),
 ('J', 'd'),
 ('Q', 'd'),
 ('K', 'd'),
 ('A', 'd')],
[('2', 'c'),
 ('3', 'c'),
 ('4', 'c'),
 ('5', 'c'),
 ('6', 'c'),
 ('7', 'c'),
 ('8', 'c'),
 ('9', 'c'),
 ('10', 'c'),
 ('J', 'c'),
 ('Q', 'c'),
 ('K', 'c'),
 ('A', 'c')]

```

```

In [285]: 1 print([list(zip(ranks, s * 13)) for s in suits])

```

```
[(('2', 's'), ('3', 's'), ('4', 's'), ('5', 's'), ('6', 's'), ('7', 's'), ('8', 's'), ('9', 's'), ('10', 's'), ('J', 's'), ('Q', 's'), ('K', 's'), ('A', 's'))], [('2', 'h'), ('3', 'h'), ('4', 'h'), ('5', 'h'), ('6', 'h'), ('7', 'h'), ('8', 'h'), ('9', 'h'), ('10', 'h'), ('J', 'h'), ('Q', 'h'), ('K', 'h'), ('A', 'h')], [('2', 'd'), ('3', 'd'), ('4', 'd'), ('5', 'd'), ('6', 'd'), ('7', 'd'), ('8', 'd'), ('9', 'd'), ('10', 'd'), ('J', 'd'), ('Q', 'd'), ('K', 'd'), ('A', 'd')], [('2', 'c'), ('3', 'c'), ('4', 'c'), ('5', 'c'), ('6', 'c'), ('7', 'c'), ('8', 'c'), ('9', 'c'), ('10', 'c'), ('J', 'c'), ('Q', 'c'), ('K', 'c'), ('A', 'c')]]
```

```
In [286]: 1 def f(list1, *, reverse = False):
2         if not reverse:
3             return sorted(list1)
4         else:
5             return sorted(list1, reverse = True)
6
7
8 print(f([3, 2, 4, 5, 1, 6, 8, 7]))
9 print(f([3, 2, 4, 5, 1, 6, 8, 7], reverse = True))
10 print(f([3, 2, 4, 5, 1, 6, 8, 7])[0])
11 print(f([3, 2, 4, 5, 1, 6, 8, 7], reverse = True)[0])
```

```
[1, 2, 3, 4, 5, 6, 7, 8]
[8, 7, 6, 5, 4, 3, 2, 1]
1
8
```

```
In [287]: 1 def f(list1, *, reverse = False):
2         return sorted(list1, reverse = reverse)
3
4
5 print(f([3, 2, 4, 5, 1, 6, 8, 7]))
6 print(f([3, 2, 4, 5, 1, 6, 8, 7], reverse = True))
7 print(f([3, 2, 4, 5, 1, 6, 8, 7])[0])
8 print(f([3, 2, 4, 5, 1, 6, 8, 7], reverse = True)[0])
```

```
[1, 2, 3, 4, 5, 6, 7, 8]
[8, 7, 6, 5, 4, 3, 2, 1]
1
8
```

```
In [288]: 1 print(hex(ord(chr(0xE3).upper())))
```

```
0xc3
```

```
In [289]: 1 print("abc" in "defabcghi")
```

```
True
```

```
In [290]: 1 print("abc" in "abbde")
```

```
False
```

```
In [291]: 1 def solve(a, b, c):
2         solution = ((-1 * b) + math.sqrt((b * b) - (4 * a * c))) / (2 * a)
3         return round(solution, 2)
4
5
```

```
6 print(solve(3, 4, -5))
7 print(solve(1, -5, -8))
```

```
0.79
6.27
```

```
In [292]: 1 def encrypt(s):
          2     return "".join(chr(ord(c) + 10) for c in s)
          3
          4
          5 print(encrypt("ABCDEFGHIJKLMNOPQRSTUVWXYZ"))
```

```
KL MN OP QR ST UV WX YZ [\]^_`abcd
```

```
In [293]: 1 def decrypt(s):
          2     return "".join(chr(ord(c) - 10) for c in s)
          3
          4
          5 print(decrypt("KL MN OP"))
          6 print(decrypt("S}kkm* Xo\x81~yx"))
```

```
ABCDEF
Isaac Newton
```

```
In [294]: 1 print(10 % 3)
```

```
1
```

```
In [295]: 1 a = array.array("l", [20])
          2 print(a)
```

```
array('l', [20])
```

```
In [296]: 1 for i in a:
          2     print(i)
```

```
20
```

```
In [297]: 1 print(sys.getsizeof(a))
```

```
88
```

```
In [298]: 1 try:
          2     print(sys.getallocatedblocks(a))
          3 except TypeError as e:
          4     print(type(e).      )
```

```
TypeError
```

```
In [299]: 1 print(sys.getsizeof(1))
```

```
28
```

```
In [300]: 1 def f(s):
          2     print(s)
          3
```

```

4
5 try:
6     f(*"hello")
7 except TypeError as ex:
8     print(type(ex).      , ex.args)

```

TypeError ('f()' takes 1 positional argument but 5 were given',)

```

In [301]: 1 def func(s):
2           print(s)
3
4
5 def apply(func, *str):
6     result = func(*str)
7     return result
8
9
10 apply(func, "john")

```

john

```

In [302]: 1 try:
2           apply(func, "john", "smith")
3 except TypeError as ex:
4     print(type(ex).      , *ex.args)

```

TypeError func() takes 1 positional argument but 2 were given

```

In [303]: 1 def encloseGetPower(b):
2           print("hex(id(b)):", hex(id(b)))
3
4           def getPower(e):
5               print("hex(id(e)):", hex(id(e)))
6
7               return b ** e
8
9           print("getPower:", getPower)
10
11          return getPower
12
13
14 getBinaryPower = encloseGetPower(2)
15 print(getBinaryPower(31))
16 getTertiaryPower = encloseGetPower(3)
17 print(getTertiaryPower(4))

```

```

hex(id(b)): 0xad7e90
getPower: <function encloseGetPower.<locals>.getPower at 0x74efc644bd70>
hex(id(e)): 0xad8230
2147483648
hex(id(b)): 0xad7eb0
getPower: <function encloseGetPower.<locals>.getPower at 0x74efc644be20>
hex(id(e)): 0xad7ed0
81

```

```
In [304]: 1 print("getBinaryPower:", getBinaryPower)
          2 print("encloseGetPower:", encloseGetPower)
          3 try:
          4     print("getPower:", getPower)
          5 except NameError as ex:
          6     print(type(ex)., ex.args)
```

```
getBinaryPower: <function encloseGetPower.<locals>.getPower at 0x74efc644bd70>
encloseGetPower: <function encloseGetPower at 0x74efc644b5e0>
NameError ("name 'getPower' is not defined",)
```

```
In [305]: 1 print("getBinaryPower:", getBinaryPower.      )
          2 print("encloseGetPower:", encloseGetPower.      )
```

```
getBinaryPower: (<cell at 0x74efc24f1510: int object at 0xad7e90>,)
encloseGetPower: None
```

```
In [306]: 1 def outer(a, b):
          2     print(hex(id(a)))
          3     sumTotal = a + b
          4
          5     def inner():
          6         prod = a * b
          7         print(a, b, sumTotal, prod)
          8         return "You just called a closure!"
          9
         10     return inner
         11
         12
         13 f = outer(2, 3)
```

```
0xad7e90
```

```
In [307]: 1 f = outer(2, 3)
```

```
0xad7e90
```

```
In [308]: 1 print(f)
```

```
<function outer.<locals>.inner at 0x74efc644bed0>
```

```
In [309]: 1 print(f.      )
```

```
(<cell at 0x74efc641f430: int object at 0xad7e90>, <cell at 0x74effaaf9db0: int object at 0xad7eb0>,
<cell at 0x74effb46b430: int object at 0xad7ef0>)
```

```
In [310]: 1 f()
```

```
2 3 5 6
```

```
Out[310]: 'You just called a closure!'
```

```
In [311]: 1 def execute(func):
          2
          3     def inner(a, b):
```

```
4         return func(a, b)
5
6     return inner
7
8
9 def add(a, b):
10     return a + b
11
12
13 add_executor = execute(add)
14 print(add_executor.)
```

(<cell at 0x74efc24f0a30: function object at 0x74efc24f8040>,)

In [312]: 1 print(hex(id(add)))

0x74efc24f8040

In [313]: 1 try:
2 print(hex(add))
3 except TypeError as ex:
4 print(type(ex).)

TypeError

In [314]: 1 add_executor(2, 3)

Out[314]: 5

In [315]: 1 def f():
2 pass
3
4
5 print(f.)

None

In [316]: 1 def f():
2
3 def g():
4 return
5
6 return g
7
8
9 h = f()
10 print(f.)
11 print(h.)

None

None

In [317]: 1 a = 0
2 try:

```
3     print(a.          )
4 except AttributeError as e:
5     print(type(e).    )
```

AttributeError

```
In [318]: 1 a = []
          2 try:
          3     print(a.          )
          4 except AttributeError as e:
          5     print(type(e).    )
```

AttributeError

```
In [319]: 1 def execute(func):
          2
          3     def inner(*posargs, **kwargs):
          4         return func(*posargs, **kwargs)
          5
          6     return inner
          7
          8
          9 def add(a, b, c):
          10     return a + b + c
          11
          12
          13 def say_hello(name, *, formal = True):
          14     return f"Pleased to meet you, {name}" if formal else f"Hi, {name}!"
          15
          16
          17 exec_add = execute(add)
          18 exec_greet = execute(say_hello)
```

```
In [320]: 1 exec_add(1, 2, 3)
```

Out[320]: 6

```
In [321]: 1 exec_greet("Michael", formal = False)
```

Out[321]: 'Hi, Michael!'

```
In [322]: 1 exec_greet("Michael", formal = True)
```

Out[322]: 'Pleased to meet you, Michael'

```
In [323]: 1 print((0.1).as_integer_ratio())
```

(3602879701896397, 36028797018963968)

```
In [324]: 1 print((0.125).as_integer_ratio())
```

(1, 8)

```
In [325]: 1 print((0.2).as_integer_ratio())
          2 print(f"{0.2:.40f}")
```

```
(3602879701896397, 18014398509481984)
0.2000000000000000111022302462515654042363
```

```
In [326]: 1 print(math.pi.as_integer_ratio())
          2 n, d = math.pi.as_integer_ratio()
          3 print(n / d)
```

```
(884279719003555, 281474976710656)
3.141592653589793
```

```
In [327]: 1 try:
          2     a1 = a1 + 1
          3     print(a1)
          4 except NameError as ex:
          5     print(type(ex).      , ": ", *ex.args, ".", sep = "")
```

```
NameError: name 'a1' is not defined.
```

```
In [328]: 1 def getDiagonalMatrix(val, rows, columns):
          2     diagonalMatrix = [[val if ce == r else 0 for ce in range(4)] for r in
          3     range(5)]
          4     return diagonalMatrix
```

```
In [329]: 1 def getDiagonalMatrix(rows, columns, *, val):
          2     diagonalMatrix = [[val if cl == row else 0 for cl in range(columns)]
          3     for row in range(rows)]
          4     return diagonalMatrix
```

```
In [330]: 1 def getFactorial(n):
          2     factorial = 1
          3     for i in range(1, n + 1):
          4         factorial *= i
          5     return factorial
```

```
In [331]: 1 result = getFactorial(5)
          2 print(result)
```

```
120
```

```
In [332]: 1 result = getDiagonalMatrix(3, 4, val = 5)
          2 pprint.pprint(result, width = 14)
```

```
[[5, 0, 0, 0],
 [0, 5, 0, 0],
 [0, 0, 5, 0]]
```

```
In [333]: 1 print(f"{timeit.timeit('getFactorial(2)', globals = globals()):.4g}")
```

```
0.1758
```

```
In [334]: 1 n = 0.000000000000123456789
2 n = 4.482882e-05
3 exp = 0
4 while n < 1:
5     exp -= 3
6     n *= 1000
7 prefixes = "", "m", "\N{GREEK SMALL LETTER MU}", "n", "p", "f", "a"
8 print(f"{n}{prefixes[abs(exp)//3]}")
```

44.82882μ

```
In [335]: 1 n = 1000000000
2 exp = 0
3 while n >= 1000:
4     exp += 3
5     n /= 1000
6 prefixes = "", "k", "M", "G", "T", "P", "E"
7 print(f"{n}{prefixes[exp//3]}")
```

1.0G

```
In [336]: def toEnggNotation(n, scientificNotation = True, nSignificantDigits = 4):
1     exp = 0
2     oldN = n
3     if n > 1:
4         while n >= 1000:
5             exp += 3
6             n /= 1000
7         prefixes = "", "k", "M", "G", "T", "P", "E"
8         return f"{n:.3f}{prefixes[exp//3]}\t{oldN}" if
scientificNotation else f"{n}{prefixes[exp//3]}"
9     elif n < 1:
10        while n < 1:
11            exp -= 3
12            n *= 1000
13        prefixes = "", "m", "\N{GREEK SMALL LETTER MU}", "n", "p", "f",
14        "a"
15        return f"{n:.{nSignificantDigits}f}
16        {prefixes[abs(exp)//3]}\t{oldN}" if scientificNotation else f"{n}
17        {prefixes[abs(exp)//3]}"
18    else:
19        return 1
```

```
1 def myTimer(f, *pos, **keyword):
2     start = time.perf_counter()
3     print(f(*pos, **keyword))
4     end = time.perf_counter()
5     return end - start
6
7
```

```
8 print(toEnggNotation(myTimer(getFactorial, 1000)))
9 print(toEnggNotation(myTimer(getDiagonalMatrix, 3, 5, val = 7)))
```

[illegible]

```

1 def encloseMyTimer(f):
2
3     def myTimer(*pos, **keyword):
4         start = time.perf_counter()
5         result = f(*pos, **keyword)
6         end = time.perf_counter()
7         print(toEnggNotation(end - start, True, 4))
8         return result
9
10    return myTimer
11
12
13 timed = encloseMyTimer(getDiagonalMatrix)
14 print(timed(2, 4, val = -5))
15 timed = encloseMyTimer(getFactorial)
16 print(timed(11))
17 print(encloseMyTimer(getDiagonalMatrix)(2, 4, val = -5))
18 print(encloseMyTimer(getFactorial)(11))

```

```
5.2100µ 5.209999983435409e-06
[[-5, 0, 0, 0], [0, -5, 0, 0]]
4.1080µ 4.108000041469495e-06
39916800
3.6170µ 3.6170000043966866e-06
[[-5, 0, 0, 0], [0, -5, 0, 0]]
2.4240µ 2.4240000016106933e-06
39916800
```

```
In [341]: 1 # print(eval("2 * 3"))
          2 # print(eval("x = 2 and print(x)"))
```

```
In [342]: 1 # eval("x = 2")
          2 # print(x)
```

```
In [343]: 1 x = 2
          2 eval("x ** 2 - 1")
```

```
Out[343]: 3
```

```
In [344]: 1 def f(expr, x):
          2     print(eval(expr))
          3
          4
          5 x = 2
          6 f("x ** 2 - 1", x)
```

```
3
```

```
In [345]: 1 def f(x):
          2     return True
          3
          4
          5 try:
          6     print(list(map(f(False), range(0, 5))))
          7 except TypeError as ex:
          8     print(type(ex).      + ":", ex.args[0])
```

```
TypeError: 'bool' object is not callable
```

```
In [346]: 1 def getExtremum(expr, interval, granularity, minimax):
          2
          3     def f(expr):
          4
          5         def g(x):
          6             return eval(expr)
          7
          8         return g
          9
          10     evalExpr = f(expr)
          11     print("Range:", list(range(interval[0], interval[1] + granularity,
          12 granularity)))
          13     if minimax == "minimum":
```

```

7         return min(map(evalExpr, range(interval[0], interval[1] +
granularity, granularity)))
8     elif minimax == "maximum":
9         return max(map(evalExpr, range(interval[0], interval[1] +
granularity, granularity)))
10    else:
11        return "Invalid minimax"
12
13    expr = "x ** 2 - 1"
14    print(getExtremum(expr, (-10, 10), 2, "minimum"))
15    print(getExtremum(expr, (-10, 10), 2, "maximum"))
16
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96
97
98
99

```

```

Range: [-10, -8, -6, -4, -2, 0, 2, 4, 6, 8, 10]
-1
Range: [-10, -8, -6, -4, -2, 0, 2, 4, 6, 8, 10]
99

```

```

In [347]: 1 expr = "x ** 2 - 1"
          2 f = lambda x: eval(expr)
          3 print(f(3))

```

8

```

In [348]: 1 def getExtremum(expr, interval, resolution, minimax):
          2
          3     def f(expr):
          4         def g(x):
          5             return eval(expr)
          6
          7         return g
          8
          9     evalExpr = f(expr)

```

```

    l = [interval[0] + (i * (interval[1] - interval[0]) / (resolution -
6    1)) for i in range(resolution)] # Cheated with the calc.
        return min(map(evalExpr, l)) if minimax == "minimum" else
7    max(map(evalExpr, l))

8
    expr = "x ** 2 - 1"
9    print(getExtremum(expr, (-10, 10), 5, "minimum"))
1    print(getExtremum(expr, (-10, 10), 5, "maximum"))
0    print(getExtremum(expr, (-10, 10), 6, "minimum"))
1    print(getExtremum(expr, (-10, 10), 6, "maximum"))
1    expr = "math.log10(x)"
1    print(getExtremum(expr, (1, 100), 20, "minimum"))
2    print(getExtremum(expr, (1, 100), 20, "maximum"))
1    expr = "math.sin(x)"
3    print(getExtremum(expr, (-10, 10), 20, "minimum"))
1    print(getExtremum(expr, (-10, 10), 20000, "maximum"))
4    expr = "x"
1    print(getExtremum(expr, (1, 6), 8, "minimum"))
5    print(getExtremum(expr, (1, 6), 8, "maximum"))

1
6
1
7
1
8
1
9
2
0
2
1
2
2
2
3
2
4
2
5
2
6
2
7
2
8

```

```
-1.0
99.0
3.0
99.0
0.0
2.0
-0.9999667804441291
0.999999996044897
1.0
6.0
```

In [349]:

```
def f(t):
    return t[0] ** t[1]

l = [(1, 1), (1, 2), (2, 1), (2, 2), (2, 3), (3, 2), (3, 3), (3, 4), (4, 3),
      (4, 4)]

start = time.perf_counter()
for _ in range(10_000):
    result = []
    for t in l:
        result.append(f(t))
end = time.perf_counter()
forLoopTime = end - start
print(result)

start = time.perf_counter()
for _ in range(10_000):
    result = [f(t) for t in l]
end = time.perf_counter()
listCompTime = end - start
print(result)

start = time.perf_counter()
for _ in range(10_000):
    result = list(map(f, l))
end = time.perf_counter()
mapTime = end - start
print(result)

print(toEnggNotation(forLoopTime), toEnggNotation(listCompTime),
      toEnggNotation(mapTime))
```

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```
[1, 1, 2, 4, 8, 9, 27, 81, 64, 256]
[1, 1, 2, 4, 8, 9, 27, 81, 64, 256]
[1, 1, 2, 4, 8, 9, 27, 81, 64, 256]
15.1299m      0.0151298600000001827  10.1019m   0.010101901999973961  10.2137m   0.010213654999972732
```

In [350]:

```
def power(x, y):
    return x ** y

def dist(pt1, pt2):
    return math.sqrt(sum(coord1 - coord2 for coord1, coord2 in zip(pt1,
pt2)))

def partial(f, operand2):

    def calcPartialFunction(operand1):
        return f(operand1, operand2)

    return calcPartialFunction

squares = partial(power, 2)
print(squares(3))
print(squares(4))
dist_from_origin = partial(dist, (0, 0))
print(dist_from_origin((1, 1)))
gcd_60 = partial(math.gcd, 60)
print(gcd_60(13))
log_2 = partial(math.log, 2)
print(log_2(8))
log_10 = partial(math.log, 10)
print(log_10(100))
```

```
5 log_16 = partial(math.log, 16)
6 print(log_16(65536))
```

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3.0
2.0
4.0
```

In [351]:

```
1 def norm(x, y):
2     return math.sqrt(x ** 2 + y ** 2)
3
4
5 def find_index_min(seq):
6     minima = min(seq)
7     return seq.index(minima)
8
9
10 def logged(f):
11
12     def g(*args):
13         print("Logging...")
14         start = time.perf_counter()
15         print(f(*args))
```

```

16         end = time.perf_counter()
17         print(toEnggNotation(end - start))
18         print("... Finished")
19
20     return g
21
22
23 loggedNorm = logged(norm)
24 loggedNorm(3, 4)
25 loggedFindIndexMin = logged(find_index_min)
26 loggedFindIndexMin([8, 7, 2, 9, 4])

```

```

Logging...
5.0
16.9020µ      1.6901999970286852e-05
... Finished
Logging...
2
12.1130µ      1.2113000025237852e-05
... Finished

```

```

In [352]: 1 def encloseGetPower(f):
2
3           def getPower(*args):
4               return f(*args)
5
6           print(hex(id(getPower)))
7           return getPower
8
9
10 getPower = encloseGetPower(pow)
11 print(getPower(2, 5))
12 print(hex(id(getPower)))
13 print(hex(id(encloseGetPower)))
14 print(getPower.          )
15 print(encloseGetPower.   )

```

```

0x74efc24f9430
32
0x74efc24f9430
0x74efc24f94e0
(<cell at 0x74efc5818ac0: builtin_function_or_method object at 0x74f01fbb0630>,)
None

```

```

In [353]: 1 l = [(1, 2), (-4, -5.5), (10, -10), (-2, 2)]
2         print(sorted(l, key = lambda t: abs(t[0] - t[1]), reverse = True))

```

```

[(10, -10), (-2, 2), (-4, -5.5), (1, 2)]

```

```

In [354]: 1 l = ["s", "0", "1"]
2         card = "".join(reversed(l))
3         print(card)
4         card = ""
5         for k in reversed(l):

```

```
6     card += k
7     print(card)
```

```
10s
10s
```

In [355]:

```
1
2
3
4
5 suits = 'shdc' # Spades, Hearts, Diamonds, Clubs
6 ranks = list('23456789') + ['10', 'J', 'Q', 'K', 'A']
7 cards = [[rank + suit for rank in ranks] for suit in suits]
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```
[['As', 'Ks', 'Qs', 'Js', '10s', '9s', '8s', '7s', '6s', '5s', '4s', '3s', '2s'], ['Ah', 'Kh', 'Qh', 'Jh', '10h', '9h', '8h', '7h', '6h', '5h', '4h', '3h', '2h'], ['Ad', 'Kd', 'Qd', 'Jd', '10d', '9d', '8d', '7d', '6d', '5d', '4d', '3d', '2d'], ['Ac', 'Kc', 'Qc', 'Jc', '10c', '9c', '8c', '7c', '6c', '5c', '4c', '3c', '2c']]
[['2s', '3s', '4s', '5s', '6s', '7s', '8s', '9s', '10s', 'Js', 'Qs', 'Ks', 'As'], ['2h', '3h', '4h', '5h', '6h', '7h', '8h', '9h', '10h', 'Jh', 'Qh', 'Kh', 'Ah'], ['2d', '3d', '4d', '5d', '6d', '7d', '8d', '9d', '10d', 'Jd', 'Qd', 'Kd', 'Ad'], ['2c', '3c', '4c', '5c', '6c', '7c', '8c', '9c', '10c', 'Jc', 'Qc', 'Kc', 'Ac']]
```

In [356]:

```
1 data = {'S1': (100, 200, 80, 180), 'S2': (10, 20, 8, 18), 'S3': (50, 150, 50, 150)}
```

```

2
3 def findMinima(data, reverse = False):
4     minimax = min if reverse else max
5     return minimax(data, key = lambda d: data[d][1] - data[d][2])
6
7
8
9
10 print(findMinima(data, reverse = True))
11 print(findMinima(data))
12

```

S2
S1

In [357]:

```

1 quotes = [('AACC', 6.05, 6.07, 6.03, 6.05, 65800), ('AAME', 1.7, 1.82, 1.7,
2 1.82, 4300), ('AAON', 24.98, 25.07, 24.9, 24.94, 28200), ('AAPL', 317.99,
3 319.57, 316.75, 317.13, 12901800), ('AATI', 3.82, 3.82, 3.74, 3.79, 194600),
4 ('AAWW', 60.89, 61.44, 60.5, 61.19, 272800), ('AAXJ', 65.4, 65.71, 65.28,
5 65.56, 390300), ('ABAT', 4.01, 4.01, 3.95, 3.99, 656300), ('ABAX', 25.26,
6 25.49, 25.04, 25.42, 73700), ('ABBC', 11.75, 11.88, 11.48, 11.53, 29700),
7 ('ABCB', 9.3, 9.3, 9.06, 9.14, 42600), ('ABCD', 3.25, 3.25, 3.11, 3.22,
8 122800), ('ABCO', 48.75, 50.41, 46.9, 50.37, 66300), ('ABCW', 0.52, 0.61, 0.52,
9 0.53, 83000), ('ABFS', 25.98, 26.27, 25.41, 25.5, 384900), ('ABIO', 3.96, 4,
10 3.88, 4, 38500), ('ABMD', 11.94, 12, 11.69, 11.87, 122600), ('ABTL', 0.82,
11 0.84, 0.82, 0.83, 28700), ('ABVA', 3.09, 3.25, 3.09, 3.25, 6200), ('ACAD',
12 0.76, 0.76, 0.7, 0.74, 341500), ('ACAS', 7.52, 7.72, 7.52, 7.66, 5199800),
13 ('ACAT', 14.44, 14.44, 14.04, 14.2, 51700), ('ACCL', 8.11, 8.21, 7.94, 8.1,
14 456100), ('ACET', 8.01, 8.04, 7.13, 7.73, 575600), ('ACFC', 1.69, 1.7, 1.5,
15 1.6, 12300), ('ACFN', 3.82, 4, 3.82, 3.98, 53700), ('ACGL', 89.76, 90.14,
16 89.39, 89.92, 240900), ('ACGY', 22.41, 22.56, 22.25, 22.46, 86800), ('ACHN',
17 3.12, 3.2, 3.07, 3.16, 113700), ('ACIW', 26.96, 27.03, 26.63, 26.8, 157000),
18 ('ACLI', 33.65, 33.77, 33.45, 33.63, 28700), ('ACLS', 2.47, 2.63, 2.46, 2.53,
19 1818800), ('ACMR', 2.69, 2.84, 2.37, 2.71, 158600), ('ACOM', 25.2, 26.6, 24.9,
20 26.56, 265300), ('ACOR', 26.67, 27.07, 26.38, 27.04, 1415000), ('ACPW', 1.84,
21 1.89, 1.77, 1.85, 565500), ('ACTG', 27.2, 27.43, 26.86, 27.18, 228800),
22 ('ACTI', 3.25, 3.26, 3.25, 3.26, 148500), ('ACTS', 2.08, 2.09, 2.07, 2.07,
23 130500), ('ACUR', 2.6, 2.64, 2.51, 2.6, 16000), ('ACWI', 46.53, 46.7, 46.32,
24 46.51, 286200), ('ACWX', 44.49, 44.66, 44.36, 44.6, 55500), ('ACXM', 18, 18.07,
25 17.81, 18.01, 289800), ('ADAM', 7.34, 7.49, 7.33, 7.44, 81700), ('ADAT', 0.6,
26 0.68, 0.59, 0.66, 86400), ('ADBE', 29.43, 29.71, 29.07, 29.14, 7585300),
27 ('ADCT', 12.68, 12.69, 12.66, 12.68, 1660500), ('ADEP', 6.14, 6.14, 4.95, 5.61,
28 71000), ('ADES', 6.2, 6.22, 6, 6.19, 4800), ('ADGF', 4.31, 4.55, 4.31, 4.54,
29 10200)]

```

```
def myFilter(percentage = 10):
    return list(filter(lambda t: abs((t[4] - t[2]) / t[2]) > (percentage / 100), quotes))

print("10:", myFilter(10))
print("5:", myFilter(5))
print("1:", myFilter(1))
```

```
10: [('ABCW', 0.52, 0.61, 0.52, 0.53, 83000)]
5: [('ABCW', 0.52, 0.61, 0.52, 0.53, 83000), ('ACFC', 1.69, 1.7, 1.5, 1.6, 12300), ('ADEP', 6.14, 6.14, 4.95, 5.61, 71000)]
1: [('ABBC', 11.75, 11.88, 11.48, 11.53, 29700), ('ABCB', 9.3, 9.3, 9.06, 9.14, 42600), ('ABCW', 0.52, 0.61, 0.52, 0.53, 83000), ('ABFS', 25.98, 26.27, 25.41, 25.5, 384900), ('ABMD', 11.94, 12, 11.69, 11.87, 122600), ('ABTL', 0.82, 0.84, 0.82, 0.83, 28700), ('ACAD', 0.76, 0.76, 0.7, 0.74, 341500), ('ACAT', 14.44, 14.44, 14.04, 14.2, 51700), ('ACCL', 8.11, 8.21, 7.94, 8.1, 456100), ('ACET', 8.01, 8.04, 7.13, 7.73, 575600), ('ACFC', 1.69, 1.7, 1.5, 1.6, 12300), ('ACHN', 3.12, 3.2, 3.07, 3.16, 113700), ('ACLS', 2.47, 2.63, 2.46, 2.53, 1818800), ('ACMR', 2.69, 2.84, 2.37, 2.71, 158600), ('ACPW', 1.84, 1.89, 1.77, 1.85, 565500), ('ACUR', 2.6, 2.64, 2.51, 2.6, 16000), ('ADAT', 0.6, 0.68, 0.59, 0.66, 86400), ('ADBE', 29.43, 29.71, 29.07, 29.14, 7585300), ('ADEP', 6.14, 6.14, 4.95, 5.61, 71000)]
```

```
In [358]: 1 l = [5, 6, -4, 8]
          2 print(min(l, key = abs))
```

```
-4
```

```
In [462]: 1 importlib.reload(logging)
          2 logging.basicConfig(format = "%(asctime)s %(levelname)s %(message)s", datefmt = "%FT%T")
          3 logging.error("This is an error")
```

```
2026-09-18T20:52:36 ERROR This is an error
```

```
In [360]: 1 logger = logging.getLogger(
          2 logger.setLevel(logging.DEBUG)
          3 logger.error("This is an error message")
          4 logger.debug("This is a debug message")
```

```
2026-09-18T17:28:00b ERROR This is an error message
2026-09-18T17:28:00b DEBUG This is a debug message
```

```
In [361]: 1 def f():
          2     pass
          3
          4
          5 print(f.
          6 print(type(f.))
```

```
f
<class 'str'>
```

```
In [362]: 1 def partial(f):
          2
          3     def callF(*pargs, **kwargs):
          4         logger.debug("Calling your function... " + f. )
```

```

5         start = time.perf_counter()
6         result = f(*pargs, **kwargs)
7         end = time.perf_counter()
8         logger.debug("... Function returned.")
9         logger.info(f"Elapsed Time: {toEnggNotation(end - start)}")
10        # print(f"{toEnggNotation(end - start)}")
11        return result
12
13    return callF
14
15
16    def add(a, b):
17        return a + b
18
19
20    def log3(a):
21        return math.log(a, 3)
22
23
24    @partial
25    def log4(a):
26        return math.log(a, 4)
27
28
29    myAdder = partial(add)
30    logger.info("myAdder: " + str(myAdder(2, 3)))
31    log3 = partial(log3)
32    logger.info("log3: " + str(log3(9)))
33    logger.info("log4: " + str(log4(16)))

```

```

2026-09-18T17:28:00b DEBUG Calling your function... add
2026-09-18T17:28:00b DEBUG ... Function returned.
2026-09-18T17:28:00b INFO Elapsed Time: 1.5930µ 1.5930000358821417e-06
2026-09-18T17:28:00b INFO myAdder: 5
2026-09-18T17:28:00b DEBUG Calling your function... log3
2026-09-18T17:28:00b DEBUG ... Function returned.
2026-09-18T17:28:00b INFO Elapsed Time: 20.6290µ 2.0628999948257842e-05
2026-09-18T17:28:00b INFO log3: 2.0
2026-09-18T17:28:00b DEBUG Calling your function... log4
2026-09-18T17:28:00b DEBUG ... Function returned.
2026-09-18T17:28:00b INFO Elapsed Time: 4.6590µ 4.659000012452452e-06
2026-09-18T17:28:00b INFO log4: 2.0

```

In [363]:

```

1    def cache(f):
2        print("Initializing cache...")
3        cacheDict = {}
4
5        def callF(*pargs):
6            if pargs in cacheDict:
7                print("Cache Hit")
8                return cacheDict[pargs]
9            else:
10               print("Cache Miss")
11               result = f(*pargs)

```

```

12         cacheDict[pargs] = result
13         return result
14
15     return callF
16
17
18 @cache
19 def add(a, b):
20     return a + b
21
22
23 @cache
24 def mul(a, b):
25     return a * b

```

Initializing cache...

Initializing cache...

```

In [364]: 1 print(add.      )
          2 print(mul.    )

```

(<cell at 0x74efc252cdf0: dict object at 0x74efc253fb00>, <cell at 0x74f01f194070: function object at 0x74efc2537b60>)

(<cell at 0x74efc252e310: dict object at 0x74efc253fb40>, <cell at 0x74efc252f220: function object at 0x74efc2537d70>)

```

In [365]: 1 print(add(2, 3))
          2 print(mul(2, 3))
          3 print(add(2, 3))
          4 print(mul(2, 3))

```

Cache Miss

5

Cache Miss

6

Cache Hit

5

Cache Hit

6

```

In [366]: 1 print(type(func tools.lru_cache))

```

<class 'function'>

```

In [367]: 1 @func tools.lru_cache(maxsize = 2)
          2 def graham(a, b, c):
          3     print("Cache Miss")
          4     return a ** b ** c
          5
          6
          7 print(gramah(2, 2, 4))

```

Cache Miss

65536

```
In [368]: 1 print(graham(2, 2, 4))
          2 print(graham(2, 2, 4))
          3 print(graham(2, 2, 5))
          4 print(graham(2, 2, 6))
          5 print(graham(2, 2, 4))
```

```
65536
65536
Cache Miss
4294967296
Cache Miss
18446744073709551616
Cache Miss
65536
```

```
In [369]: 1 @functools.lru_cache(maxsize = 100)
          2 def fib(n):
          3     print(f"Calling fib{n}...")
          4     if n <= 1:
          5         return n
          6     return fib(n - 1) + fib(n - 2)
          7
          8
          9 print(fib(2000))
```

Calling fib2000...
Calling fib1999...
Calling fib1998...
Calling fib1997...
Calling fib1996...
Calling fib1995...
Calling fib1994...
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In [370]:

```
1 def myTimer(f):
2
3     def callF(*args):
4         start = time.perf_counter()
5         result = f(*args)
```

```

6         end = time.perf_counter()
7         print(toEnggNotation(end - start))
8         return result
9
10        return callF
11
12
13    # timedPower = myTimer(math.pow)
14    # print(timedPower(5, 188))
15    #
16    #
17    @myTimer
18    def myPower(a, b):
19        return a ** b
20
21
22    print(myPower(5, 188))

```

```

2.5650µ 2.5650000452515087e-06
2548947057811923643246208636428338888945767288308190057718217365417860642420893057508943799552292297
50557351508177816867828369140625

```

```
In [371]: 1 PRECISION = 4
```

```
In [372]: 1 a = decimal.Decimal(math.e)
2 print(a)
3 print(type(a))
4 print(float(a))
```

```

2.718281828459045090795598298427648842334747314453125
<class 'decimal.Decimal'>
2.718281828459045

```

```
In [373]: 1 a = decimal.Decimal("2.404040313893824388847472828")
2 print(a)
3 print(type(a))
4 print(float(a))
```

```

2.404040313893824388847472828
<class 'decimal.Decimal'>
2.4040403138938244

```

```
In [374]: 1 a = decimal.Decimal(2.2384892948592929275758693)
2 print(a)
3 print(type(a))
4 print(float(a))
```

```

2.23848929485929293292656439007259905338287353515625
<class 'decimal.Decimal'>
2.238489294859293

```

```
In [375]: 1 l = [decimal.Decimal(random.random()) for _ in range(10)]
2 a = sum(l) / len(l)
```

```
3 print(a)
4 print(type(a))
```

```
0.6355070068397963400919081777
<class 'decimal.Decimal'>
```

In [376]:

```
1 def normalize(f):
2
3     def callF(*args):
4         result = f(*args)
5         # print(type(result))
6         result = float(result)
7         return float(f"{result:.{PRECISION}f}")
8
9     return callF
```

In [377]:

```
1 @normalize
2 def perc_diff(x, y):
3     try:
4         return (y - x) / x * 100
5     except ZeroDivisionError:
6         return 0
7
8
9 @normalize
10 def sum_squares(*args):
11     return sum(x ** 2 for x in args)
12
13
14 @normalize
15 def avg(*args):
16     if len(args) == 0:
17         return 0
18
19     numbers = [decimal.Decimal(x) for x in args]
20     sum_ = sum(numbers)
21     return sum_ / len(numbers)
```

In [378]:

```
1 perc_diff(5088, 2872)
```

Out[378]: -43.5535

In [379]:

```
1 avg(2.4, 4.1, 5.932929, 8.2838485)
```

Out[379]: 5.1792

In [380]:

```
1 def add(x, y):
2     time.sleep(2)
3     return x + y
4
```

```
5 print(timeit.timeit("add(2340320832580385043843805805,  
6 320342408320482804320584305)", globals = globals(), number = 10))
```

20.000953659999993

In [381]:

```
1 @functools.lru_cache  
2 def add(x, y):  
3     time.sleep(2)  
4     return x + y  
5  
6  
7 print(timeit.timeit("add(2340320832580385043843805805,  
320342408320482804320584305)", globals = globals(), number = 10))
```

2.0001189919999547

In [382]:

```
1 def add(x, y):  
2     time.sleep(2)  
3     return x + y  
4  
5  
6 print(timeit.repeat("add(2340320832580385043843805805,  
320342408320482804320584305)", globals = globals(), number = 10))
```

[20.000912009000047, 20.000890174000006, 20.000836774999982, 20.00079347799999, 20.000815604000024]

In [383]:

```
1 @functools.lru_cache  
2 def add(x, y):  
3     time.sleep(2)  
4     return x + y  
5  
6  
7 print(timeit.repeat("add(2340320832580385043843805805,  
320342408320482804320584305)", globals = globals(), number = 10))
```

[2.000094929999989, 1.943999961895315e-06, 1.5530000041508174e-06, 1.4529999816659256e-06,
1.4830000054644188e-06]

In [384]:

```
1 def graham(x, y, z):  
2     return x ** y ** z  
3  
4  
5 print(toEnggNotation(timeit.timeit("graham(4, 5, 6)", globals = globals(),  
number = 10)))
```

686.5070μ 0.0006865070000117157

In [385]:

```
1 @functools.lru_cache  
2 def graham(x, y, z):  
3     return x ** y ** z  
4  
5  
6 print(toEnggNotation(timeit.timeit("graham(2, 3, 4)", globals = globals(),  
number = 10)))
```

3.9570μ 3.9569999898958486e-06

```
In [386]: 1 print(toEnggNotation(timeit.timeit("2 ** 3 ** 4", number = 100)))
```

18.8660μ 1.8866000004891248e-05

```
In [387]: 1 def normalize(precision):
2
3     def callF(f):
4
5         def callFInner(*args):
6             result = f(*args)
7             result = f"{result:.{precision}f}"
8             return result
9
10        return callFInner
11
12    return callF
```

```
In [388]: 1 @normalize(2)
2 def perc_diff(x, y):
3     try:
4         return (y - x) / x * 100
5     except ZeroDivisionError:
6         return 0
7
8
9 @normalize(4)
10 def sum_squares(*args):
11     return sum(x ** 2 for x in args)
12
13
14 @normalize(8)
15 def avg(*args):
16     if len(args) == 0:
17         return 0
18
19     numbers = [decimal.Decimal(x) for x in args]
20     sum_ = sum(numbers)
21     return sum_ / len(numbers)
```

```
In [389]: 1 print(perc_diff(3, 7))
2 print(sum_squares(3, 4, 5))
3 print(avg(1.25, 1.75, 1.5, 1.0, 2.0, 2.5, 2.25, 2.75))
```

133.33
50.0000
1.87500000

```
In [390]: 1 def normalize(f):
2
3     def callF(*args):
```

```

4         result = f(*args)
5         return result
6
7     return callF

```

In [391]:

```

1  try:
2
3      @normalize(2)
4      def perc_diff(x, y):
5          try:
6              return (y - x) / x * 100
7          except ZeroDivisionError:
8              return 0
9
10     @normalize(4)
11     def sum_squares(*args):
12         return sum(x ** 2 for x in args)
13
14     @normalize(8)
15     def avg(*args):
16         if len(args) == 0:
17             return 0
18
19         numbers = [decimal.Decimal(x) for x in args]
20         sum_ = sum(numbers)
21         return sum_ / len(numbers)
22
23 except TypeError as ex:
24     print(type(ex).      + ": " + ex.args[0] + ".")

```

TypeError: 'int' object is not callable.

In [392]:

```

@functools.lru_cache
1 def getFibonacci(i):
2     if i == 0:
3         return 1
4     if i == 1:
5         return 1
6     if i < 0:
7         return 0
8     else:
9         return getFibonacci(i - 1) + getFibonacci(i - 2)
10
11 print(getFibonacci(60))
12
13 with open("../var/out000.txt", "w") as fl:
14     fl.write("i,fib,fact,gcd\n")
15     for i in range(101):
16         fl.write(f"{i},{math.factorial(i)},{getFibonacci(i)},
17 {math.gcd(getFibonacci(i),math.factorial(i))}\n")
18
19 0

```

```
1 with open("../var/out000.txt") as fl:
1   for line in fl:
1     print(line.strip())
2
1
3
1
4
1
5
1
6
1
7
1
8
1
9
2
0
2
1
2
2
```

2504730781961
i,fib,fact,gcd
0,1,1,1
1,1,1,1
2,2,2,2
3,6,3,3
4,24,5,1
5,120,8,8
6,720,13,1
7,5040,21,21
8,40320,34,2
9,362880,55,5
10,3628800,89,1
11,39916800,144,144
12,479001600,233,1
13,6227020800,377,13
14,87178291200,610,10
15,1307674368000,987,21
16,20922789888000,1597,1
17,355687428096000,2584,136
18,6402373705728000,4181,1
19,121645100408832000,6765,165
20,2432902008176640000,10946,26
21,51090942171709440000,17711,1
22,112400072777607680000,28657,1
23,25852016738884976640000,46368,46368
24,620448401733239439360000,75025,25
25,1551121004330985984000000,121393,1
26,403291461126605635584000000,196418,34
27,10888869450418352160768000000,317811,39
28,304888344611713860501504000000,514229,1
29,8841761993739701954543616000000,832040,440
30,265252859812191058636308480000000,1346269,1
31,8222838654177922817725562880000000,2178309,21
32,263130836933693530167218012160000000,3524578,2
33,8683317618811886495518194401280000000,5702887,1
34,295232799039604140847618609643520000000,9227465,65
35,10333147966386144929666651337523200000000,14930352,139536
36,371993326789901217467999448150835200000000,24157817,1
37,13763753091226345046315979581580902400000000,39088169,37
38,52302261746660111760007224100074291200000000,63245986,2
39,20397882081197443358640281739902897356800000000,102334155,1155
40,815915283247897734345611269596115894272000000000,165580141,1
41,33452526613163807108170062053440751665152000000000,267914296,3016
42,1405006117752879898543142606244511569936384000000000,433494437,1
43,60415263063373835637355132068513997507264512000000000,701408733,129
44,2658271574788448768043625811014615890319638528000000000,1134903170,170
45,119622220865480194561963161495657715064383733760000000000,1836311903,1
46,5502622159812088949850305428800254892961651752960000000000,2971215073,1
47,258623241511168180642964355153611979969197632389120000000000,4807526976,4358592
48,12413915592536072670862289047373375038521486354677760000000000,7778742049,13
49,608281864034267560872252163321295376887552831379210240000000000,12586269025,275
50,3041409320171337804361260816606476884437764156896051200000000000,20365011074,2
51,155111875328738228022424301646930321106325972001698611200000000000,32951280099,3
52,8065817517094387857166063685640376697528950544088327782400000000000,53316291173,1
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54,23084369733924138047209274268302758108327856457180794113228800000000000,139583862445,5
55,1269640335365827592596510084756651695958032105144943676227584000000000000,225851433717,55419
56,71099858780486345185404564746372494973649797888116845868744704000000000000,365435296162,74
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59,13868311854568983573793901972038940634590287677268743254082129494016000000000000,1548008755920,1
0066320
60,83209871127413901442763411832233643807541726063612459524492776964096000000000000,2504730781961,
1
61,5075802138772247988008568121766252272260045289880360030994059394809856000000000000,405273953788
1,1
62,3146997326038793752565312235495076408801228079725823219216316824782110720000000000000,6557470319
842,442
63,198260831540444006411614670836189813754477369022726862810627959961272975360000000000000,10610209
857723,987
64,12688693218588416410343338933516148080286551617454519219880189437521470423040000000000000,171676
80177565,5
65,824765059208247066672317030678549625218625855134543749292212313438895577497600000000000000,27777
890035288,8
66,54434493907744306400372924024784275264429306438879887453286012686967108114841600000000000000,449
45570212853,1
67,3647111091818868528824985909660546442716763531404952459370162850026796243694387200000000000000,7
2723460248141,201
68,248003554243683059960099041856917158104739920135536767237171073801822144571218329600000000000000
,117669030460994,2
69,1711224524281413113724683388812728390922705448935203693936480409232572797541406474240000000000000
00,190392490709135,20735
70,119785716699698917960727837216890987364589381425464258575536286462800958278984531968000000000000
0000,308061521170129,1
71,8504785885678623175211676442399260102885846081207962358864307633885886803780790176972800000000000
00000,498454011879264,44930592
72,6123445837688608686152407038527467274077809178469732898382301496397838498722168927420416000000000
0000000,806515533049393,1
73,44701154615126843408912571381250511100768007002829050158190800923704221040671833170169036800000000
000000000,1304969544928657,73
74,3307885441519386412259530282212537821456832518209349711706119268354112357009715654592508723200000
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0000000000000,5527939700884757,13
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79,8946182130782975286851441715398316520698082167795719072138680632278379906935018605333618108410101
7600000000000000000,23416728348467685,2225685
80,7156945704626380229481153372318653216558465734236575257710944505822703925548014884266894486728081
408000000000000000000,37889062373143906,1802
81,5797126020747367985879734231578109105412357244731625958745865049716390179693892056256184534249745
94048000000000000000000,61305790721611591,1
82,4753643337012841748421382069894049466438132940679933286171609340767439947348991486130071318084791
67119360000000000000000000,99194853094755497,1
83,3945523969720658651189747118012061057143650340764344627522435752836975156299662933487959194010377
087090688000000000000000000,160500643816367088,4505904
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7531561779200000000000000000,259695496911122585,5
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240182751232000000000000000000,420196140727489673,1
86,2422709538367273238176552320344125971528487055242938175083876449672016224974245027678946463490131
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7935047344717824000000000000000000,1100087778366101931,903
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018284166335168512000000000000000000,1779979416004714189,1
89,1650795516090846108121691926245361930983966623649654185491352070783317103437850973939991257078760

```

066272908038299975680000000000000000000,2880067194370816120,268748920
90,1485715964481761497309522733620825737885569961284688766942216863704985393094065876545992131370884
05964561723446997811200000000000000000000,4660046610375530309,169
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4942775116833676800819200000000000000000000,7540113804746346429,3
92,1243841405464130725547532432587355307757799171587541435684023958293813771098351951844304612383704
134735310748698265675366400000000000000000000,12200160415121876738,2
93,1156772507081641574759205162306240436214753229576413535186142281213246807121467315215203289516844
84530383899628938707809075200000000000000000000,19740274219868223167,1
94,1087366156656743080273652852567866010041868035801828723074973744340451998694179276302291092145834
1545856086565120238534053068800000000000000000000,31940434634990099905,185
95,1032997848823905926259970209939472709539774634011737286921225057123429398759470312487176537538542
44685632822368642266073504153600000000000000000000,51680708854858323072,8717184
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26512612841896796781676470678323200000000000000000000,135301852344706746049,36569
98,942689044888324774562618574305724247380969376407895166349423877294707070023223798882976159207729
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100,933262154439441526816992388562667004907159682643816214685929638952175999932299156089414639761565
1828625369792082722375825118521091686400000000000000000000,573147844013817084101,1

```

In [393]:

```

fibonacciNumbers = {}
1 factorialNumbers = {}
gcd = {}
2
with open("../var/out000.txt") as fl:
3     next(fl)
    for ln in fl:
4         l = ln.strip().split(",")
        factorialNumbers[int(l[0])] = int(l[1])
5         fibonacciNumbers[int(l[0])] = int(l[2])
        gcd[(int(l[1]), int(l[2]))] = int(l[3])
6
7 # @functools.lru_cache
def getFibonacci(i):
8     print(f"getFibonacci({i}):")
    if i < 0:
9         return 0
    if i in fibonacciNumbers:
1        print("Found in cache!")
        return fibonacciNumbers[i]
1        print("Not found in cache...")
    fibonacciNumbers[i] = getFibonacci(i - 1) + getFibonacci(i - 2)
2        return fibonacciNumbers[i]
1
3
1 print(getFibonacci(110))
4 print(getFibonacci(11))
1
5
1 def f(i):

```

```

6
1      def g():
7          return math.factorial(i)
1
8      return g
1
9
2  def getFactorial(i):
0      print(f"getFactorial({i}):")
2      if i in factorialNumbers:
1          print("Found in cache!")
2          return factorialNumbers[i]
2      print("Not found in cache...")
2      return f(i)()
3          # return factorialNumbers.get(i, f(i)())      # Doesn't work because
2          the closure is always called. Have to use condition.
4
2
5  print(getFactorial(110))
2  print(getFactorial(11))
6
2
7  def f1(i, j):
2
8      def g1():
2          return math.gcd(i, j)
9
3      return g1
0
3
1  def getGCD(i, j):
3      print(f"getGCD({i, j}):")
2      if (i, j) in gcd:
3          print("Found in cache!")
3          return gcd[(i, j)]
3      print("Not found in cache...")
4      return math.gcd(i, j)
3
5
3  print(getGCD(5040, 21))
6  print(getGCD(5040, 22))
3
7
3
8
3
9
4
0
4
1
4

```

2
4
3
4
4
4
5
4
6
4
7
4
8
4
9
5
0
5
1
5
2
5
3
5
4
5
5
5
6
5
7
5
8
5
9
6
0
6
1
6
2
6
3
6
4
6
5
6
6
6
7
6

8
6
9
7
0
7
1

[illegible]

```
In [394]: 1 def importMPL():
2         import matplotlib as mpl
3
4         return mpl
5
6
7 mpl = importMPL()
8 print(dir(mpl))
```

```
['ExecutableNotFoundError', 'MatplotlibDeprecationWarning', 'MutableMapping', 'Parameter', 'Path', 'RcParams', '_ExecInfo', '_VersionInfo', '__all__', '__bibtex__', '__builtins__', '__cached__', '__doc__', '__file__', '__getattr__', '__loader__', '__name__', '__package__', '__path__', '__spec__', '_add_data_doc', '_api', '_c_internal_utils', '_check_versions', '_cm', '_cm_bivar', '_cm_listed', '_cm_multivar', '_color_data', '_deprecated_ignore_map', '_deprecated_map', '_deprecated_remain_as_none', '_docstring', '_ensure_handler', '_enums', '_fontconfig_pattern', '_get_config_or_cache_dir', '_get_executable_info', '_get_ssl_context', '_get_version', '_get_xdg_cache_dir', '_get_xdg_config_dir', '_image', '_init_tests', '_label_from_arg', '_log', '_logged_cached', '_open_file_or_url', '_parse_to_version_info', '_path', '_preprocess_data', '_rc_params_in_file', '_replacer', '_val_or_rc', '_version', 'artist', 'atexit', 'backends', 'bezier', 'bivar_colormaps', 'cbook', 'cm', 'color_sequences', 'colorizer', 'colormaps', 'colors', 'contextlib', 'cycler', 'defaultParams', 'ft2font', 'functools', 'get_backend', 'get_cachedir', 'get_configdir', 'get_data_path', 'importlib', 'inspect', 'interactive', 'is_interactive', 'locale', 'logging', 'matplotlib_fname', 'multivar_colormaps', 'namedtuple', 'os', 'parse_version', 'path', 'pprint', 'rc', 'rcParams', 'rcParamsDefault', 'rcParamsOrig', 'rc_context', 'rc_file', 'rc_file_defaults', 'rc_params', 'rc_params_from_file', 'rcdefaults', 'rcsetup', 're', 'sanitize_sequence', 'scale', 'set_loglevel', 'shutil', 'subprocess', 'sys', 'tempfile', 'ticker', 'transforms', 'use', 'validate_backend']
```

```
In [395]: 1 epochTime = time.gmtime(0)
2         print(epochTime.tm_year, epochTime.tm_mday, epochTime.tm_hour,
epochTime.tm_min, epochTime.tm_zone)
```

```
1970 1 0 0 GMT
```

```
In [396]: 1 time.perf_counter()
```

```
Out[396]: 440.340111055
```

```
In [397]: 1 start = time.perf_counter()
2         time.sleep(3)
3         end = time.perf_counter()
4         print(end - start)
```

```
3.0002507969999783
```

```
In [398]: 1 print(time.strftime("%I:%M %p %z", time.gmtime(time.time())))
```

```
12:00 PM +0000
```

```
In [ ]: 1 print(datetime.datetime.now(datetime.UTC))
2        print(datetime.datetime.now())
```

```
2026-09-18 12:00:08.720941+00:00
2026-09-18 17:30:08.721215
```

```
In [400]: 1 print(time.strftime("%Y-%m-%dT%H:%M:%S%z", time.gmtime(time.time())))
```

2026-09-18T12:00:08+0000

```
In [401]: my_time_struct = time.strptime("2020-09-17T16:36:16Z", "%Y-%m-%dT%H:%M:%S%z")
1 print(my_time_struct.tm_year, my_time_struct.tm_mon, my_time_struct.tm_mday,
2 my_time_struct.tm_hour, my_time_struct.tm_min, my_time_struct.tm_sec,
my_time_struct.tm_gmtoff)
```

2020 9 17 16 36 16 0

```
In [402]: my_time_struct = time.strptime("2020-09-17T16:36:16+0530", "%Y-%m-%dT%H:%M:%S%z")
1
2 print(my_time_struct.tm_year, my_time_struct.tm_mon, my_time_struct.tm_mday,
my_time_struct.tm_hour, my_time_struct.tm_min, my_time_struct.tm_sec,
my_time_struct.tm_gmtoff)
```

2020 9 17 16 36 16 19800

```
In [403]: 1 print("hello")
```

hello

```
In [404]: 1 s = "Monday, April 18, 2020"
2 myTimeStruct = time.strptime(s, "%A, %B %d, %Y")
3 print(myTimeStruct)
```

time.struct_time(tm_year=2020, tm_mon=4, tm_mday=18, tm_hour=0, tm_min=0, tm_sec=0, tm_wday=0, tm_yday=109, tm_isdst=-1)

```
In [420]: 1 myDate = datetime.date(2020, 8, 20)
2 print(myDate)
3 print(repr(myDate))
4 print(myDate.day, myDate.month)
```

2020-08-20
datetime.date(2020, 8, 20)
20 8

```
In [419]: 1 print(myDate.isoformat())
```

2020-08-20

```
In [421]: 1 myDate = datetime.date.today()
2 print(myDate)
3 print(repr(myDate))
4 print(myDate.day, myDate.month)
```

2026-09-18
datetime.date(2026, 9, 18)
18 9

```
In [ ]: 1 print(myDate.isoformat())
```

2026-09-18
18 9

```
In [407]: 1 myDate = datetime.date.fromtimestamp(time.time())
          2 print(myDate.year)
```

2026

```
In [408]: 1 myDate = datetime.date.fromtimestamp(0)
          2 print(myDate.year)
```

1970

```
In [ ]: 1 myDate = datetime.date.fromisoformat("2020-08-24")
        2 print(myDate)
        3 print(repr(myDate))
```

2020-08-24

datetime.date(2020, 8, 24)

```
In [410]: 1 myTime = datetime.time(15, 23, 45)
          2 print(myTime.isoformat())
```

15:23:45

```
In [411]: 1 print(datetime.time(15, 30, 45, 600))
```

15:30:45.000600

```
In [424]: 1 try:
          2     myTime = datetime.time.now()
          3     print(myTime)
          4 except AttributeError as ex:
          5     print(type(ex).__name__, ex.args)
```

AttributeError ("type object 'datetime.time' has no attribute 'now'",)

```
In [412]: 1 print(datetime.time.fromisoformat("12:34:45.123"))
```

12:34:45.123000

```
In [413]: 1 myTime = datetime.time.fromisoformat("12:34:45.1234")
          2 print(myTime)
          3 print(myTime.hour)
          4 print(myTime.microsecond)
          5 try:
          6     print(myTime[:3])
          7 except TypeError as ex:
          8     print(type(ex).__name__, ex.args)
```

12:34:45.123400

12

123400

TypeError ("'datetime.time' object is not subscriptable",)

```
In [423]: 1 try:
          2     myTime = datetime.time.fromtimestamp(0)
          3     print(myTime)
```

```
4 except AttributeError as ex:
5     print(type(ex)., ex.args)
```

AttributeError ("type object 'datetime.time' has no attribute 'fromtimestamp',)

```
In [414]: 1 myDateTime = datetime.datetime(2020, 12, 9, 12, 43, 32, 38829)
2         print(myDateTime.isoformat())
```

2020-12-09T12:43:32.038829

```
In [426]: 1 myDateTime = datetime.datetime.now()
2         print(myDateTime)
3         print(repr(myDateTime))
4         print(myDateTime.hour)
5         print(myDateTime.isoformat())
```

2026-09-18 18:03:08.224995
datetime.datetime(2026, 9, 18, 18, 3, 8, 224995)
18
2026-09-18T18:03:08.224995

```
In [429]: 1 myDateTime = datetime.datetime.fromtimestamp(0)
2         print(myDateTime)
3         print(myDateTime.hour)
4         print(myDateTime.isoformat())
```

1970-01-01 05:30:00
5
1970-01-01T05:30:00

```
In [436]: 1 myDateTime = datetime.datetime.fromtimestamp(time.time())
2         print(myDateTime)
3         print(myDateTime.hour)
4         print(myDateTime.isoformat())
```

2026-09-18 18:11:43.908956
18
2026-09-18T18:11:43.908956

```
In [431]: 1 myDateTime = datetime.datetime.fromisoformat("2025-02-02T15:23:44+0545")
2         print(myDateTime)
3         print(myDateTime.hour)
4         print(myDateTime.isoformat())
```

2025-02-02 15:23:44+05:45
15
2025-02-02T15:23:44+05:45

```
In [432]: 1 myDateTime = datetime.datetime.fromisoformat("2025-02-02T15:23:44")
2         print(myDateTime)
3         print(myDateTime.hour)
4         print(myDateTime.isoformat())
```

2025-02-02 15:23:44
15
2025-02-02T15:23:44

```
In [433]: 1 myDateTime = datetime.datetime.fromisoformat("2025-02-02 15:23:44")
          2 print(myDateTime)
          3 print(myDateTime.hour)
          4 print(myDateTime.isoformat())
```

```
2025-02-02 15:23:44
15
2025-02-02T15:23:44
```

```
In [434]: 1 myDateTime = datetime.datetime.fromisoformat("2025-02-02 15:23")
          2 print(myDateTime)
          3 print(myDateTime.hour)
          4 print(myDateTime.isoformat())
```

```
2025-02-02 15:23:00
15
2025-02-02T15:23:00
```

```
In [435]: 1 myDateTime = datetime.datetime.fromisoformat("2025-02-02")
          2 print(myDateTime)
          3 print(myDateTime.hour)
          4 print(myDateTime.isoformat())
```

```
2025-02-02 00:00:00
0
2025-02-02T00:00:00
```

```
In [438]: 1 myDateTime = datetime.datetime.utcnow()
          2 print(myDateTime)
          3 print(myDateTime.hour)
          4 print(myDateTime.tzinfo)
          5 print(myDateTime.isoformat())
```

```
2026-09-18 12:54:43.306385
12
None
2026-09-18T12:54:43.306385

/tmp/ipykernel_18021/573860576.py:1: DeprecationWarning: datetime.datetime.utcnow() is deprecated
and scheduled for removal in a future version. Use timezone-aware objects to represent datetimes in
UTC: datetime.datetime.now(datetime.UTC).
  myDateTime = datetime.datetime.utcnow()
```

```
In [440]: 1 myDateTime = datetime.datetime.now(datetime.UTC)
          2 print(myDateTime)
          3 print(myDateTime.hour)
          4 print(myDateTime.tzinfo)
          5 print(myDateTime.isoformat())
```

```
2026-09-18 12:55:53.446540+00:00
12
UTC
2026-09-18T12:55:53.446540+00:00
```

```
In [442]: 1 try:
          2     myDateTime = datetime.datetime.now(datetime.IST)
          3     print(myDateTime)
```

```

4     print(myDateTime.hour)
5     print(myDateTime.tzinfo)
6     print(myDateTime.isoformat())
7 except AttributeError as ex:
8     print(type(ex).      , ex.args)

```

AttributeError ("module 'datetime' has no attribute 'IST',)

In [415]: 1 print(cmath.sqrt(-4))

2j

In [451]:

```

1 dt1 = datetime.datetime.now()
2 print(dt1)
3 dt2 = datetime.datetime.today()
4 print(dt2)
5 td = dt1 - dt2
6 print(toEnggNotation(abs(td.total_seconds()))))

```

2026-09-18 18:53:23.132215
 2026-09-18 18:53:23.132384
 169.00000μ 0.000169

In [459]:

```

1 dt1 = datetime.datetime.now()
2 print(dt1)
3 startOfDay = datetime.datetime.fromisoformat(datetime.date.today().isoformat())
4 print(startOfDay)
5 td = dt1 - startOfDay
6 print(td.total_seconds())

```

2026-09-18 18:57:44.890444
 2026-09-18 00:00:00
 68264.890444

In [461]:

```

1 td = datetime.timedelta(weeks = 52, seconds = 86400)
2 print(datetime.datetime.now() + td)

```

2027-09-18 19:01:37.580957

In [467]:

```

1 print(time.strftime("%FT%T", time.strptime("1/6/2024 2:30 PM", "%d/%m/%Y %I:%M %p")))

```

2024-06-01T14:30:00

In [472]:

```

1 td = datetime.timedelta(days = -1, seconds = 5400)
2 print(td + datetime.datetime.now())
3 print(td.total_seconds() // 60 / 60)

```

2026-09-18 18:41:55.925042
 -22.5

In [480]:

```

1 myTZ = datetime.timezone(datetime.timedelta(hours = 5.5), name = "IST")
2 print(myTZ)
3 UTCTZ = datetime.timezone.utc
4 print(UTCTZ)

```

IST
UTC

In [489]:

```
1 print(datetime.datetime.now().replace(tzinfo =  
datetime.timezone(datetime.timedelta(hours = 5.5), name =  
"IST")).astimezone(datetime.timezone(datetime.timedelta(hours =  
9))).isoformat())
```

2026-09-19T21:28:11.050467+09:00

In [507]:

```
1 currentTime = datetime.datetime.now()  
2 tzIST = datetime.timezone(datetime.timedelta(hours = 5.5))  
3 currentISTTime = currentTime.replace(tzinfo = tzIST)  
4 tzJST = datetime.timezone(datetime.timedelta(hours = 9))  
5 currentJSTTime = currentISTTime.astimezone(tzJST)  
6 print(currentJSTTime)  
7 currentJSTTime = currentJSTTime.replace(tzinfo = None)  
8 tzDifference = currentJSTTime - datetime.datetime.now()  
9 print(tzDifference)  
10 print(type(tzDifference).      )  
11 print(round(tzDifference.total_seconds() // 60 / 60, 1))
```

2026-09-19 21:49:02.698258+09:00
3:29:59.999810
timedelta
3.5

In [514]:

```
1 epochTime = 12345678.9  
2 dt = datetime.datetime.fromtimestamp(epochTime).replace(day = 1, hour = 0,  
3 minute = 0, second = 0, microsecond = 0)  
print(repr(dt))
```

datetime.datetime(1970, 5, 1, 0, 0)

In []:

```
1 epochTime = 12345678.9  
2 dt = datetime.datetime.fromtimestamp(epochTime)  
3 print(repr(datetime.datetime(year = dt.year, month = dt.month, day = 1)))
```

datetime.datetime(2026, 9, 1, 0, 0)

In []:

```
1 epochTime = time.time()  
2  
3 dt = datetime.datetime.fromtimestamp(epochTime)  
4 print(repr(datetime.datetime(year = dt.year, month = dt.month, day = 1)))
```

datetime.datetime(2026, 9, 1, 0, 0)

In [520]:

```
1 start = "2001-01-01T13:50:23"  
2 end = "2001-06-12T14:23:50"  
3 diff = int(round((datetime.datetime.fromisoformat(end) -  
4 datetime.datetime.fromisoformat(start)).total_seconds() // 60 / 60, 0))  
print(diff)
```

3889

In [544]:

```
1 start = "2001-01-01T00:00:00"
2 end = "2001-01-01T01:00:00"
3 diff = int(round((datetime.datetime.fromisoformat(end) -
4 datetime.datetime.fromisoformat(start)).total_seconds() // 60 / 60, 0))
5 print(diff)
```

1

In [545]:

```
1 start = "2001-01-01T00:00:00"
2 end = "2001-01-02T01:50:00"
3 diff = int(round((datetime.datetime.fromisoformat(end) -
4 datetime.datetime.fromisoformat(start)).total_seconds() // 60 / 60, 0))
5 print(diff)
```

26

In [534]:

```
1 def formatDateTime(dateTimeObj, formatStr = "%FT%T"):
2     return dateTimeObj.strftime(formatStr)
3
4
5 dateTimeObj = datetime.datetime.strptime("02/04/2025 03:30:30 PM", "%m/%d/%Y
6 %I:%M:%S %p")
7 print(formatDateTime(dateTimeObj))
8 print(formatDateTime(dateTimeObj, "%Y-%m-%d"))
```

2025-02-04T15:30:30

2025-02-04

In [546]:

```
1 def formatDateTime(dateTimeObj, formatStr = "%FT%T"):
2     return dateTimeObj.strftime(formatStr)
3
4
5 dateTimeObj = datetime.datetime(2020, 2, 1, 13, 34, 5)
6 print(formatDateTime(dateTimeObj))
7 print(formatDateTime(dateTimeObj, "%Y-%m-%d"))
```

2020-02-01T13:34:05

2020-02-01

In []:

```
1 with open("../var/actors.csv") as fh:
2     itr = csv.reader(fh)
3     for row in itr:
4         print(*row)
```

First Name Last Name DOB Sketches

John Cleese 10/27/39 The Cheese Shop, Ministry of Silly Walks, It's the Arts

Eric Idle 3/29/43 The Cheese Shop, Nudge Nudge, "Spam"

Peter O'Toole 8/2/32 Lawrence of Arabia

In []:

```
1 with open("../var/actors.pdv") as fh:
2     itr = csv.reader(fh, delimiter = "|", quotechar = "'", escapechar =
3     "\\ ", skipinitialspace = True)
4     for row in itr:
5         print(row)
```

```
['FIRST_NAME', 'LAST_NAME', 'DOB', 'SKETCHES']  
['John', 'Cleese', '10/27/39', "The Cheese Shop, Ministry of Silly Walks, It's the Arts"]  
['Eric', 'Idle', '3/29/43', 'The Cheese Shop, Nudge Nudge, "Spam"']  
['Peter', "O'Toole", '8/2/32', 'Lawrence of Arabia']
```

```
In [557]: 1 csv.register_dialect("pdv", delimiter = "|", quotechar = "'", skipinitialspace  
          2 = True, escapechar = "\\")  
          csv.list_dialects()
```

```
Out[557]: ['excel', 'excel-tab', 'unix', 'pdv']
```

```
In [563]: 1 with open("../var/actors.pdv") as fh:  
          2     itr = csv.reader(fh, dialect = "pdv")  
          3     for row in itr:  
          4         print(row)
```

```
['FIRST_NAME', 'LAST_NAME', 'DOB', 'SKETCHES']  
['John', 'Cleese', '10/27/39', "The Cheese Shop, Ministry of Silly Walks, It's the Arts"]  
['Eric', 'Idle', '3/29/43', 'The Cheese Shop, Nudge Nudge, "Spam"']  
['Peter', "O'Toole", '8/2/32', 'Lawrence of Arabia']
```

```
In [566]: 1 with open("../var/file1.csv") as fh:  
          2     for ln in fh:  
          3         print(ln, end = "")
```

```
col1,col2,col3  
10,20,30  
30,40,50  
60,70,80
```

```
In [571]: 1 with open("../var/file1.csv") as fh:  
          2     itr = csv.reader(fh)  
          3     smallest, largest = math.inf, -math.inf  
          4     for row in itr:  
          5         for fld in row:  
          6             try:  
          7                 n = int(fld)  
          8             except ValueError:  
          9                 pass  
         10             if n < smallest:  
         11                 smallest = n  
         12             if n > largest:  
         13                 largest = n  
         14  
         15     print(smallest, largest)
```

```
10 80
```

```
In [572]: 1 with open("../var/file2.csv") as fh:  
          2     for ln in fh:  
          3         print(ln, end = "")
```

```
1, 3.14, 500  
20, 1, -2  
-1.1, -2.2, -3.3
```

```
In [588]: 1 print(type(2.0))
          2 print(type(2.0) == float)
          3 print(isinstance(2.0, float))
```

```
<class 'float'>
True
True
```

```
In [609]: 1 a = "500.0"
          2 b = "500"
          3 if "." in a:
          4     print("a is fractional")
          5 if "." in b:
          6     print("b is fractional")
```

```
a is fractional
```

```
In [611]: 1 with open("../var/file2.csv") as fh:
          2     itr = csv.reader(fh, skipinitialspace = True)
          3     smallest, largest = math.inf, -math.inf
          4     for row in itr:
          5         for fld in row:
          6             n = float(fld) if "." in fld else int(fld)
          7             if n < smallest:
          8                 smallest = n
          9             if n > largest:
          10                 largest = n
          11
          12 print(smallest, largest)
```

```
-3.3 500
```

```
In [614]: 1 with open("../var/file3.tsv") as fh:
          2     for ln in fh:
          3         print(ln, end = "")
```

```
col1    col2    col3
10       20       30
40       50       60
100      200      300
```

```
In [623]: 1 with open("../var/file3.tsv") as fh:
          2     itr = csv.reader(fh, skipinitialspace = True, delimiter = "\t")
          3     smallest, largest = math.inf, -math.inf
          4     next(fh)
          5     for row in itr:
          6         for fld in row:
          7             n = int(fld)
          8             if n < smallest:
          9                 smallest = n
          10             if n > largest:
          11                 largest = n
```

```
12
13 print(smallest, largest)
```

```
10 300
```

```
In [627]: 1 with open("../var/file4.csv") as fh:
          2     for ln in fh:
          3         print(ln, end = "")
```

```
-col1-|-col2-|-col3-
10|20|30
-3.14-|-25-|-100-
---3.14-|20|-30-
```

```
In [633]: 1 a = "--3.3-"
          2 print(a.removesuffix("-").removeprefix("-"))
```

```
-3.3
```

```
In [ ]: 1 print(a.replace("-", ""))
```

```
3.3
```

```
In [642]: 1 with open("../var/file4.csv") as fh:
          2     itr = csv.reader(fh, delimiter = "|")
          3     smallest, largest = math.inf, -math.inf
          4     next(fh)
          5     for row in itr:
          6         for fld in row:
          7             fld = fld.replace("-", "")
          8             n = float(fld) if "." in fld else int(fld)
          9             if n < smallest:
         10                 smallest = n
         11             if n > largest:
         12                 largest = n
         13
         14 print(smallest, largest)
```

```
3.14 100
```

```
In [648]: 1 a = "b"
          2 print(ord(a) - ord("a"))
```

```
1
```

```
In [655]: 1 data = [{'a': '1_a', 'b': '1_b', 'c': '1_c'}, {'c': '2_c', 'd': '2_d'}, {'a':
          2     '3_a', 'c': '3_c', 'e': '3_e'}]
          3 with open("../var/out2.csv", "w") as fh:
          4     itr = csv.writer(fh)
          5     itr.writerow("abcde")
          6     for d in data:
          7         row = ["", "", "", "", ""]
          8         for k, v in d.items():
          9             row[ord(k) - ord("a")] = v
         10     itr.writerow(row)
```

```
6 with open("../var/out2.csv") as fh:
7     for ln in fh:
8         print(ln, end = "")
```

```
8
```

```
9
```

```
1
```

```
0
```

```
1
```

```
1
```

```
1
```

```
2
```

```
1
```

```
3
```

```
a,b,c,d,e
1_a,1_b,1_c,,
,,2_c,2_d,
3_a,,3_c,,3_e
```